

Triangles: their kinds and their elements

Three sides, three angles, and the two ways of classifying every triangle there is.

LESSON 46

1 × 40 MIN

MATEMATIKA 7, §19

S8 8.1

LESSON CLOCK

10'

12'

12'

6'

Elements and notation	10 min
Two classifications	12 min
The triangle inequality	12 min
Homework	6 min

LEARNING OBJECTIVES

- Name the elements of a triangle and use the standard notation.
- Classify triangles by their sides and by their angles.
- Use the triangle inequality to decide whether three lengths can form a triangle.
- Identify the median, the altitude and the bisector of a triangle.

TEXTBOOK

National	pp. 118–122
Cambridge	Stage 8, pp. 78–84
Workbook	Exercise 8.1

01

Explanation

Elements and notation

A triangle has three **vertices**, three **sides** and three **angles**. The standard notation names the side by the small letter of the opposite vertex.

ELEMENT	WRITTEN
vertices	A, B, C
sides	$a = BC, b = CA, c = AB$
angles	$\angle A, \angle B, \angle C$
perimeter	$P = a + b + c$

Three special segments from a vertex:

SEGMENT	GOES TO
median	the midpoint of the opposite side
altitude	the opposite side, at a right angle
bisector	the opposite side, halving the angle

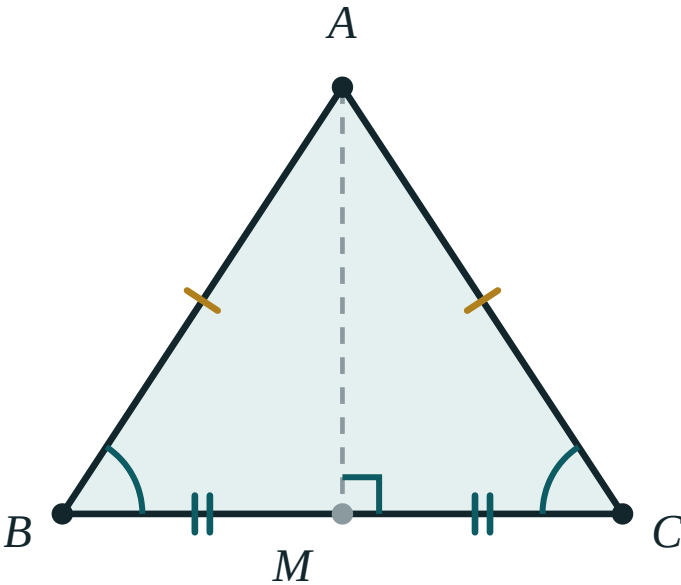
SIDE A FACES ANGLE A

That single convention makes every later formula readable — the sine rule, the cosine rule, the area formula. It is worth fixing now.

Two classifications

BY SIDES	MEANS	BY ANGLES	MEANS
scalene	all sides different	acute	all angles acute
isosceles	two sides equal	right	one angle 90°
equilateral	all three equal	obtuse	one angle obtuse

The two classifications are independent: a triangle may be isosceles and right, or scalene and obtuse, and so on. An equilateral triangle is always acute, with all angles 60°.



$AB = AC \implies \angle B = \angle C$

An isosceles triangle — two equal sides and two equal base angles.

EVERY EQUILATERAL TRIANGLE IS ISOSCELES

“Two sides equal” does not mean “exactly two”. The equilateral triangle is the special case in which all three are.

The triangle inequality

Each side is less than the sum of the other two

$$a < b + c, b < c + a, c < a + b$$

It is enough to check the **longest** side: if the longest is less than the sum of the other two, all three inequalities hold.

LENGTHS	CHECK	TRIANGLE?
3, 4, 5	$5 < 3 + 4$	yes
2, 3, 6	$6 > 2 + 3$	no
4, 4, 8	$8 = 4 + 4$	no — flat
7, 10, 15	$15 < 7 + 10$	yes

EQUALITY GIVES NO TRIANGLE

4, 4, 8 lays the two short sides flat along the long one. The inequality must be strict.

02 Worked examples

EXAMPLE 1 Can a triangle have sides 2, 3, 6? And 7, 10, 15?

- ① Longest of the first: 6.

- ② $2 + 3 = 5 < 6$ — no triangle.

- ③ Longest of the second: 15.

- ④ $7 + 10 = 17 > 15$ — yes.

ANSWER

No; yes

EXAMPLE 2 A triangle has $a = 5$ and $b = 9$. Between which values must c lie?

- ① $c < 5 + 9 = 14$

- ② $9 < 5 + c \Rightarrow c > 4$

- ③ Combine.

- ④ $4 < c < 14$

ANSWER

$4 < c < 14$

EXAMPLE 3 Classify a triangle with sides 5, 5, 8 and one angle of 106° .

- ① Two sides equal: isosceles.
By sides.
- ② One angle above 90° : obtuse.
By angles.
- ③ Both classifications apply at once.
- ④ An obtuse isosceles triangle.

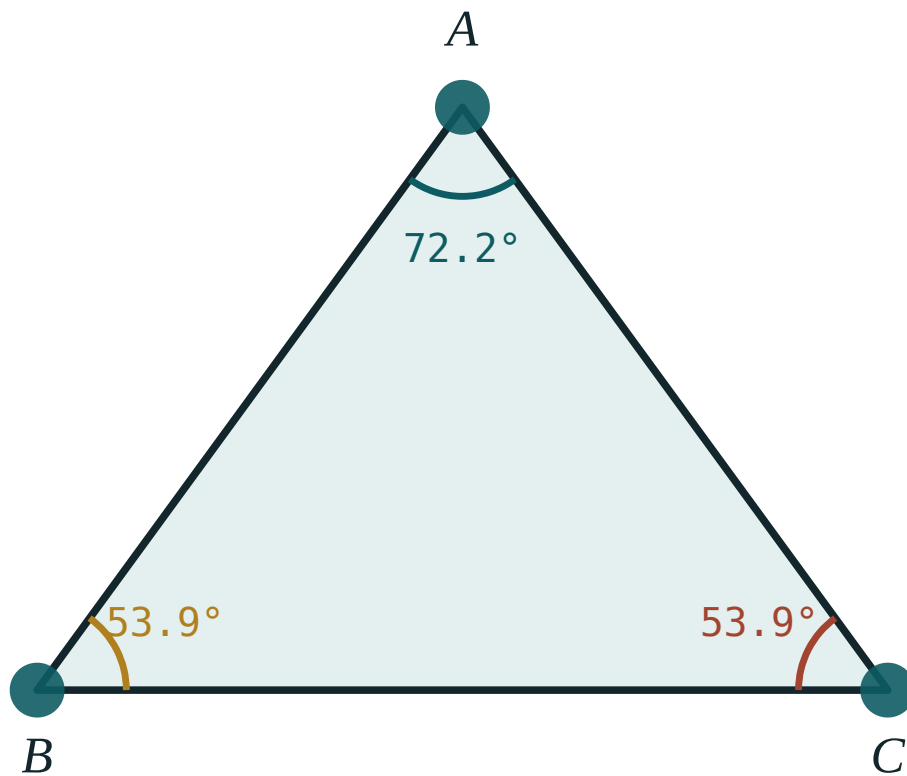
ANSWER

Obtuse isosceles

03 Interactive model

Give out three strips of card of lengths 2, 3 and 6 cm and ask the class to make a triangle; the failure is more convincing than the inequality.

Kinds of triangle

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Triangle	Uchburchak	Треугольник
Vertex	Uchi	Вершина
Side	Tomon	Сторона
Scalene	Turli tomonli	Разносторонний
Isosceles	Teng yonli	Равнобедренный
Equilateral	Teng tomonli	Равносторонний
Median	Mediana	Медиана
Altitude	Balandlik	Высота

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Side a is opposite:

A triangle with two equal sides is:

An equilateral triangle has angles:

Can 2, 3, 6 form a triangle?

yes

no

sometimes

only if right

A median goes to:

the midpoint

at a right angle

the bisected angle

the vertex

With $a = 5$, $b = 9$, c satisfies:

$c < 14$

$c > 4$

$4 < c < 14$

$c = 14$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Side opposite $\angle B$

ANSWER b

2. A triangle with all sides different

ANSWER Scalene

3. A triangle with all sides equal

ANSWER Equilateral

4. Angles of an equilateral triangle

ANSWER 60° each

5. Can 3, 4, 5 be a triangle?

ANSWER Yes

6. Can 2, 3, 6 be a triangle?

ANSWER No

7. Can 4, 4, 8 be a triangle?

ANSWER No

Medium · the standard question

7 PROBLEMS

1. Can 7, 10, 15 be a triangle?

ANSWER Yes

2. $a = 5$, $b = 9$: range of c

ANSWER $4 < c < 14$

3. $a = 6$, $b = 6$: range of c

ANSWER $0 < c < 12$

4. Classify 5, 5, 8 with a 106° angle

ANSWER Obtuse isosceles

5. Classify 3, 4, 5

ANSWER Right scalene

6. Perimeter of a triangle 7, 9, 12

ANSWER 28

7. A median goes to

ANSWER The midpoint

Hard · stretch and reason

7 PROBLEMS

1. An isosceles triangle with perimeter 20 and base 8

ANSWER Sides 6, 6, 8

2. The possible integer values of c with $a = 5$, $b = 9$

ANSWER 5 to 13

3. How many of them?

ANSWER 9

4. An isosceles triangle with sides 3 and 7

ANSWER Only 3, 7, 7

5. Why not 3, 3, 7?

ANSWER $3 + 3 < 7$

6. A triangle with perimeter 12 and integer sides: how many?

ANSWER 3

7. Name them

ANSWER 2,5,5; 3,4,5; 4,4,4

07 Homework

Homework — 5 tasks

Check the longest side against the sum of the other two, every time.

- Can a triangle have sides 4, 6, 11? And 8, 9, 16?
- A triangle has $a = 7$ and $b = 11$. Give the range of c .
- Classify a triangle with sides 6, 6, 6.
- Classify a triangle with a 90° angle and two equal sides.
- An isosceles triangle has perimeter 26 and base 10. Find the equal sides.

The first criterion of congruence of triangles

Two sides and the angle between them fix a triangle completely.

LESSON 47–48

2 × 40 MIN

MATEMATIKA 7, §20

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

What congruence means	13 min
The criterion	23 min
Writing a proof	22 min
Consequences	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define congruent triangles and use the symbol $\cong$.
- State and apply the SAS criterion.
- Write a proof of congruence in statement-and-reason form.
- Deduce equal sides and angles from a congruence.

TEXTBOOK

National	pp. 123–129
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

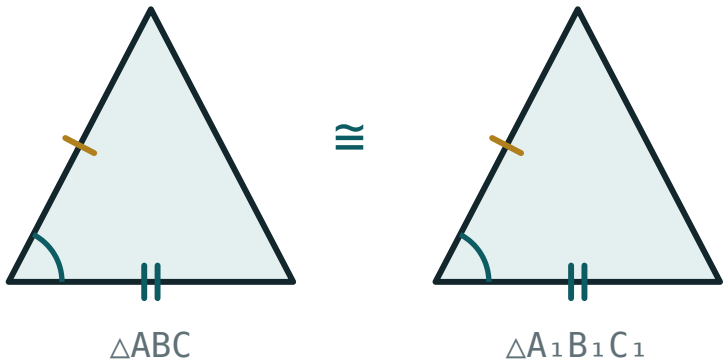
01

Explanation

What congruence means

Two triangles are **congruent** if one can be laid exactly on the other. Written $\triangle ABC \cong \triangle A_1B_1C_1$, with the vertices in matching order.

Congruent triangles have **six** pairs of equal elements: three sides and three angles.



Congruent triangles — same shape and same size, in any position.

THE ORDER OF THE LETTERS IS PART OF THE STATEMENT

$\triangle ABC \equiv \triangle A_1B_1C_1$ says A matches A_1 , B matches B_1 and C matches C_1 . Writing them in the wrong order claims something false.

The criterion

If two sides of one triangle and the angle between them are equal to two sides and the angle between them of another, the triangles are congruent

$$AB = A_1B_1, AC = A_1C_1, \angle A = \angle A_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

Three of the six pairs are enough — provided they are the **right** three.

GIVEN	CONGRUENT?
two sides and the included angle	yes — the first criterion
two sides and a non-included angle	not necessarily
three angles	no — only similar

WHY THE ANGLE MUST BE INCLUDED

Two sides fixed at a hinge open and close: the third side changes with the angle. Fixing that angle fixes the triangle. With a non-included angle the hinge is not fixed and two shapes remain possible.

Writing a proof

Problem. $AB = AD$ and $\angle BAC = \angle DAC$. Prove $\triangle ABC \equiv \triangle ADC$.

STATEMENT	REASON
$AB = AD$	given
$\angle BAC = \angle DAC$	given
$AC = AC$	common side
$\triangle ABC \equiv \triangle ADC$	first criterion (SAS)

THE COMMON SIDE IS THE MARK MOST OFTEN MISSED

Almost every proof of this kind needs it, and it is always available. Looking for a shared side or a shared angle is the first thing to do in any congruence question.

Consequences

Once a congruence is proved, **all six** pairs of elements are equal. That is what makes the criteria useful: three facts are given, and three more follow free.

PROVED	THEN ALSO
$\triangle ABC \equiv \triangle ADC$	$BC = DC$
same	$\angle B = \angle D$
same	$\angle BCA = \angle DCA$

The abbreviation used in proofs is *CPCT* — corresponding parts of congruent triangles.

MATCH THE ELEMENTS BY POSITION, NOT BY APPEARANCE

In $\triangle ABC \equiv \triangle ADC$, BC corresponds to DC because B matches D and C matches C . Reading the correspondence off the letters is always safe; reading it off the drawing is not.

02 Worked examples

EXAMPLE 1 $AB = AD$ and $\angle BAC = \angle DAC$. Prove $\triangle ABC \equiv \triangle ADC$.

- ① $AB = AD$ — given.

- ② $\angle BAC = \angle DAC$ — given.
The included angle.

- ③ AC is common to both.
The extra fact.

- ④ By the first criterion, $\triangle ABC \equiv \triangle ADC$.

ANSWER

Proved by SAS

EXAMPLE 2 In the same figure, what else follows?

- ① All six pairs of elements are equal.
CPCT.

- ② $BC = DC$

- ③ $\angle B = \angle D$

- ④ $\angle BCA = \angle DCA$

ANSWER

$BC = DC$, $\angle B = \angle D$, $\angle BCA = \angle DCA$

EXAMPLE 3 Two triangles have sides 6 and 8 with an angle of 40° . Must they be congruent?

- ① If the 40° lies between the 6 and the 8: yes.
SAS.
- ② If it lies elsewhere: not necessarily.
- ③ The position of the angle decides.
- ④ Only when it is the included angle.

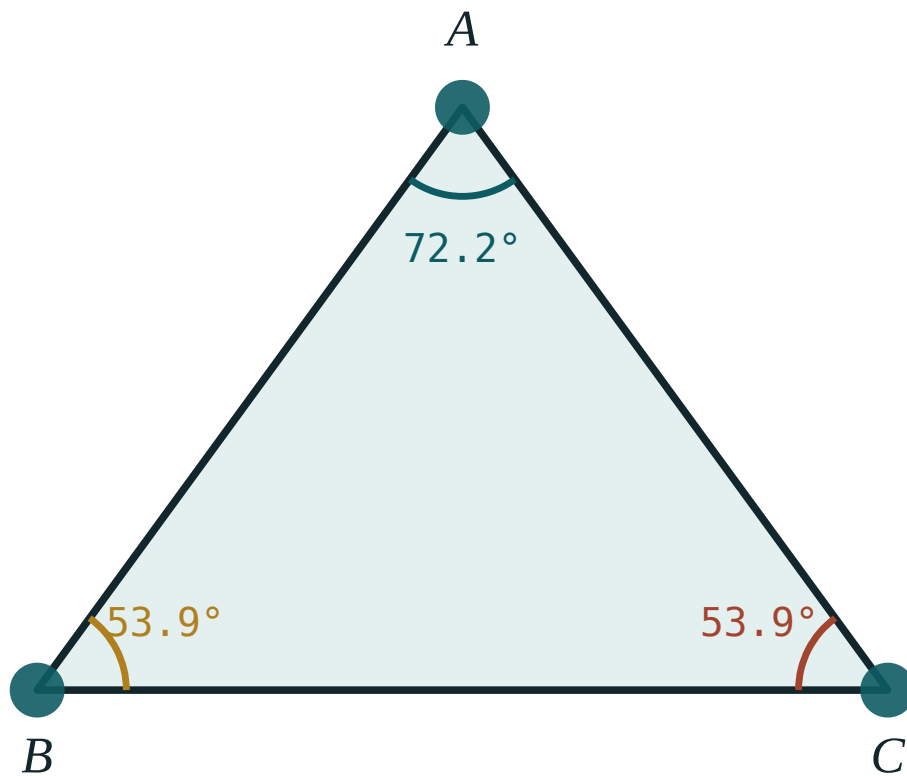
ANSWER

Only if the angle is included

03 Interactive model

Hinge two rulers at a fixed angle and mark the third side; changing the angle changes it, and the criterion becomes a physical fact.

Two sides and the angle between

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Congruent	Teng	Равный
Criterion	Alomat	Признак
Corresponding elements	Mos elementlar	Соответственные элементы
Included angle	Orasidagi burchak	Угол между сторонами
Superposition	Ustma-ust qo'yish	Наложение
To follow	Kelib chiqmoq	Следовать
Common side	Umumiy tomon	Общая сторона
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Congruent triangles have how many pairs of equal elements?

The first criterion needs:

Three equal angles give:

A shared side in a figure is:

irrelevant

a usable equality

a mistake

an angle

CPCT means:

a criterion

corresponding parts of congruent triangles

a construction

a theorem name

A non-included equal angle gives:

congruence

not necessarily congruence

similarity only

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Congruent triangles have how many equal pairs?

ANSWER 6

2. The symbol for congruence

ANSWER $\equiv$

3. The first criterion in letters

ANSWER SAS

4. A common side gives

ANSWER An equality for free

5. Three equal angles give

ANSWER Similarity only

6. In $\triangle ABC \equiv \triangle DEF$, AB matches

ANSWER DE

7. In the same, $\angle C$ matches

ANSWER $\angle F$

Medium · the standard question

7 PROBLEMS

1. $AB = AD$, $\angle BAC = \angle DAC$: the third fact

ANSWER AC common

2. What follows about BC ?

ANSWER $BC = DC$

3. What follows about $\angle B$?

ANSWER $\angle B = \angle D$

4. Sides 6, 8 with 40° included: congruent?

ANSWER Yes

5. Sides 6, 8 with 40° not included: congruent?

ANSWER Not necessarily

6. Two triangles with 5, 7 and 60° included

ANSWER Congruent

7. Are congruent triangles similar?

ANSWER Yes, with $k = 1$

Hard · stretch and reason

7 PROBLEMS

1. In a parallelogram, prove the two triangles cut by a diagonal are congruent

ANSWER SAS with the common diagonal

2. Prove that the diagonals of a parallelogram bisect each other

ANSWER Congruent triangles at the centre

3. An isosceles triangle: prove the bisector from the apex is a median

ANSWER SAS

4. Two triangles with $AB = DE$, $AC = DF$, $\angle A = \angle D$: name the criterion

ANSWER The first

5. Why is AAA not enough?

ANSWER Any enlargement has the same angles

6. In $\triangle ABC \equiv \triangle DEF$ with $AB = 5$, $\angle B = 40^\circ$: find DE and $\angle E$

ANSWER 5 and 40°

7. Prove that a kite has two pairs of congruent triangles

ANSWER SAS on each diagonal

07 Homework

Homework — 5 tasks

Look for a common side or a common angle in every figure before starting.

- In a figure with $AB = AC$ and $\angle BAD = \angle CAD$, prove $\triangle ABD \equiv \triangle ACD$.
- State what else follows from that congruence.
- Explain why the equal angle must lie between the two equal sides.
- Two triangles have sides 7 and 9 with an included angle of 55° . Are they congruent?
- Prove that a diagonal of a parallelogram divides it into two congruent triangles.

Properties of isosceles triangles

Two equal sides force two equal angles — and one line does the work of three.

LESSON 49–50

2 × 40 MIN

MATEMATIKA 7, §21

S8 5.2

LESSON CLOCK

13'

23'

22'

18'

4'

The base angles	13 min
The proof	23 min
The converse	22 min
One line, three roles	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and prove that the base angles of an isosceles triangle are equal.
- State and use the converse.
- Know that the bisector from the apex is also the median and the altitude.
- Solve numerical problems on isosceles and equilateral triangles.

TEXTBOOK

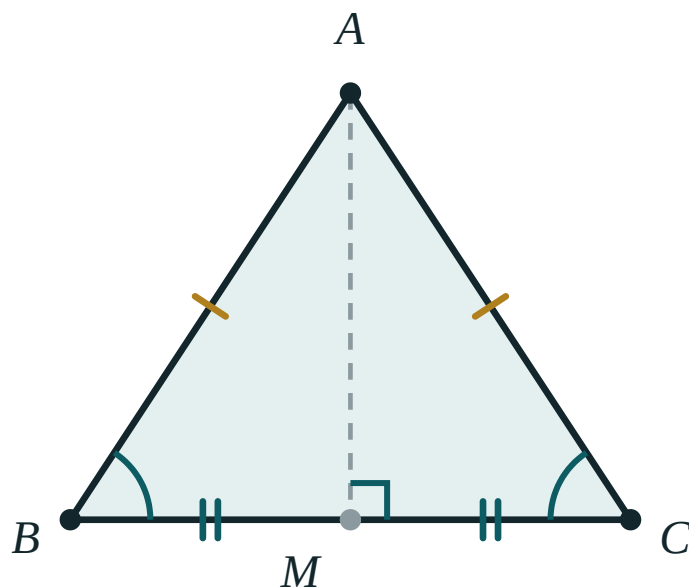
National	pp. 130–136
Cambridge	Stage 8, pp. 50–53
Workbook	Exercise 5.2

01

Explanation

The base angles

In an isosceles triangle the angles at the base are equal



$$AB = AC \Rightarrow \angle B = \angle C$$

Two equal sides, two equal base angles — and one axis of symmetry.

NAME	IN $\triangle ABC$ WITH $AB = AC$
lateral sides	AB, AC
base	BC
base angles	$\angle B, \angle C$
apex angle	$\angle A$

Since the angles add to 180° : $\angle B = \angle C = \frac{180^\circ - \angle A}{2}$, and $\angle A = 180^\circ - 2\angle B$.

ONE ANGLE DETERMINES ALL THREE

Give any one angle of an isosceles triangle — apex or base — and the other two follow at once. That is what makes these questions quick.

The proof

Draw the bisector AD of the apex angle.

STATEMENT	REASON
$AB = AC$	given
$\angle BAD = \angle CAD$	AD bisects $\angle A$
$AD = AD$	common
$\triangle ABD \equiv \triangle ACD$	first criterion
$\angle B = \angle C$	CPCT

THE AUXILIARY LINE IS THE WHOLE PROOF

Without AD there are no triangles to compare. Drawing the right extra line is the skill this proof teaches, and it recurs throughout the year.

The converse

If two angles of a triangle are equal, the sides opposite them are equal

So a triangle is isosceles **if and only if** two of its angles are equal. Either fact may be used to prove the other.

GIVEN	CONCLUDE
$AB = AC$	$\angle B = \angle C$
$\angle B = \angle C$	$AB = AC$
all three sides equal	all three angles 60°
all three angles equal	all three sides equal

A CONVERSE MUST BE PROVED SEPARATELY

That a statement is true does not make its converse true. Here both directions happen to hold, but each needs its own proof — and in general they can differ.

One line, three roles

In an isosceles triangle the bisector from the apex is also the median and the altitude to the base

It is also the axis of symmetry of the triangle. Four descriptions of one segment.

THE SEGMENT AD	BECAUSE
bisects $\angle A$	by construction
$BD = DC$	CPCT — it is a median
$AD \perp BC$	$\angle ADB = \angle ADC = 90^\circ$
is the axis of symmetry	the two halves are congruent

ONLY FROM THE APEX

The bisector from a **base** angle is not a median and not an altitude. The property belongs to the apex alone.

02 Worked examples

EXAMPLE 1 An isosceles triangle has apex angle 40° . Find the base angles.

1

The three angles add to 180° .

2

$180^\circ - 40^\circ = 140^\circ$ for the two base angles.

3

They are equal.

4

70° each.

ANSWER

70° each

EXAMPLE 2 An isosceles triangle has a base angle of 72° . Find the apex angle.

① Both base angles are 72° .

② $72 + 72 = 144$

③ $180 - 144$

④ $= 36^\circ$

ANSWER

36°

EXAMPLE 3 In $\triangle ABC$, $\angle B = \angle C = 55^\circ$ and $BC = 12$. What can be said about the sides, and about the altitude from A?

① Equal angles $\Rightarrow$ equal opposite sides: $AB = AC$.
The converse.

② So the triangle is isosceles with base BC .

③ The altitude from A is also the median.

④ It meets BC at its midpoint, 6 from each end.

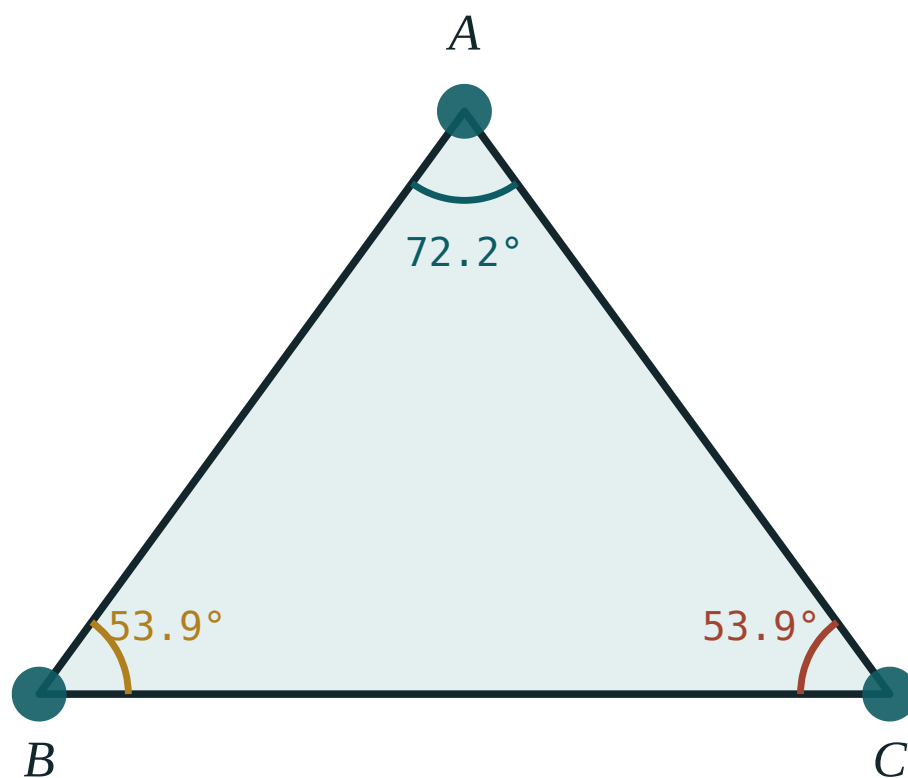
ANSWER

$AB = AC$; the altitude bisects BC

03 Interactive model

Fold a paper isosceles triangle along the line from the apex; the two halves coincide, and all four properties of that line appear at once.

Isosceles angles

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Isosceles triangle	Teng yonli uchburchak	Равнобедренный треугольник
Lateral side	Yon tomon	Боковая сторона
Base	Asos	Основание
Base angle	Asosdagi burchak	Угол при основании
Apex angle	Uchidagi burchak	Угол при вершине
Converse	Teskari teorema	Обратная теорема
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Property	Xossa	Свойство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In an isosceles triangle the equal angles are:

at the apex

at the base

all three

none

Apex 40° gives base angles:

40°

70°

140°

100°

Base angle 72° gives apex:

36°

72°

108°

144°

Two equal angles imply:

nothing

two equal sides

a right angle

equal areas

The bisector from the apex is also:

a median only

an altitude only

both

neither

An equilateral triangle has angles:

45°

60°

90°

various

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Apex 40° : base angles

ANSWER 70°

2. Apex 90° : base angles

ANSWER 45°

3. Base angle 72° : apex

ANSWER 36°

4. Base angle 50° : apex

ANSWER 80°

5. Angles of an equilateral triangle

ANSWER 60°

6. Two equal angles imply

ANSWER Two equal sides

7. The bisector from the apex is also

ANSWER A median and an altitude

Medium · the standard question

7 PROBLEMS

1. Apex 100° : base angles

ANSWER 40°

2. Base angle 35° : apex

ANSWER 110°

3. An isosceles triangle with a 90° apex: base angles

ANSWER 45°

4. An isosceles triangle with a 90° base angle: possible?

ANSWER No

5. $\angle B = \angle C = 55^\circ$: $\angle A$

ANSWER 70°

6. Perimeter 26, base 10: the lateral sides

ANSWER 8 each

7. Perimeter 30, lateral side 11: the base

ANSWER 8

Hard · stretch and reason

7 PROBLEMS

1. An isosceles triangle with base 10 and lateral side 13: its height

ANSWER 12

2. Its area

ANSWER 60

3. An isosceles triangle in which the apex is twice a base angle

ANSWER $90^\circ, 45^\circ, 45^\circ$

4. An isosceles triangle in which a base angle is twice the apex

ANSWER $36^\circ, 72^\circ, 72^\circ$

5. An exterior angle at the base of an isosceles triangle is 130° : the apex

ANSWER 80°

6. Why can an isosceles triangle not have two obtuse angles?

ANSWER The sum would exceed 180°

7. A triangle with angles $x, x, 4x$

ANSWER $30^\circ, 30^\circ, 120^\circ$

07 Homework

Homework — 5 tasks

Say which angle is the apex and which are the base angles before computing.

1. An isosceles triangle has apex angle 64° . Find the base angles.
2. An isosceles triangle has a base angle of 43° . Find the apex angle.
3. An isosceles triangle has perimeter 34 and base 12. Find the lateral sides.
4. Prove that the base angles of an isosceles triangle are equal.
5. An isosceles triangle has base 16 and lateral side 17. Find its height and area.

The second criterion of congruence of triangles

A side and the two angles on it — the criterion that surveyors use.

LESSON 51–52

2 × 40 MIN

MATEMATIKA 7, §22

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

The criterion	13 min
Which side counts	23 min
In proofs	22 min
Measuring a distance	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply the ASA criterion.
- Recognise when the given side is between the two angles.
- Use the criterion in proofs involving parallel lines.
- Apply it to measuring an inaccessible distance.

TEXTBOOK

National	pp. 137–142
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01

Explanation

The criterion

If a side and the two angles on it in one triangle equal a side and the two angles on it in another, the triangles are congruent

$$BC = B_1C_1, \angle B = \angle B_1, \angle C = \angle C_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

The third angle is then equal too, since the angles of a triangle add to 180°.

TWO ANGLES AND ANY SIDE ARE ENOUGH

Because the third angle follows from the angle sum, a side and any two angles determine the triangle — whether the side lies between them or not. That is why the criterion is sometimes written ASA and sometimes AAS.

Which side counts

GIVEN	ENOUGH?	WHY
$a, \angle B, \angle C$	yes	the side is between them
$a, \angle A, \angle B$	yes	$\angle C$ follows from the sum
$\angle A, \angle B, \angle C$	no	no side fixes the size

The middle row is the one that surprises: the side need not be included, because the missing angle is never really missing.

THREE ANGLES FIX THE SHAPE, NEVER THE SIZE

Any enlargement of a triangle has the same three angles. At least one length is always needed.

In proofs

Problem. $AB \parallel CD$ and $AB = CD$. Prove $\triangle ABO \equiv \triangle DCO$, where O is the intersection of AD and BC .

STATEMENT	REASON
$AB = CD$	given
$\angle ABO = \angle DCO$	alternate angles, $AB \parallel CD$
$\angle BAO = \angle CDO$	alternate angles
$\triangle ABO \equiv \triangle DCO$	second criterion

PARALLEL LINES ARE A SUPPLY OF EQUAL ANGLES

Almost every use of the second criterion begins with a pair of parallels. Spotting them turns a hard figure into an easy proof.

Measuring a distance

To measure the width of a river without crossing it:

1. mark A on one bank and a landmark B on the other;
2. walk along the bank to C and measure AC ;
3. measure $\angle BAC$ and $\angle BCA$;

4. construct a triangle with the same side and two angles on level ground, and measure its third side.

By the second criterion the constructed triangle is congruent to the real one, so the measured side **is** the width of the river.

THIS IS WHAT SURVEYING WAS, BEFORE TRIGONOMETRY

A tape and a protractor, a criterion of congruence, and a drawing on level ground. The method is ancient, exact, and still taught for exactly that reason.

02 Worked examples

EXAMPLE 1 Two triangles have a side of 7 with angles 50° and 60° on it. Are they congruent?

- ① A side and the two angles on it.
- ② The second criterion applies.
- ③ The third angle is 70° in both.
- ④ Congruent.

ANSWER

Yes, by ASA

EXAMPLE 2 Two triangles have a side of 7, an angle of 50° on it and an angle of 70° opposite it. Congruent?

- 1 The third angle is $180 - 50 - 70 = 60^\circ$.
- 2 So all three angles are known in both.
- 3 With the side of 7 in the same position, ASA applies.
- 4 Congruent.
The side need not be included.

ANSWER

Yes

EXAMPLE 3 $AB \parallel CD$ and $AB = CD$. Prove that AD and BC bisect each other.

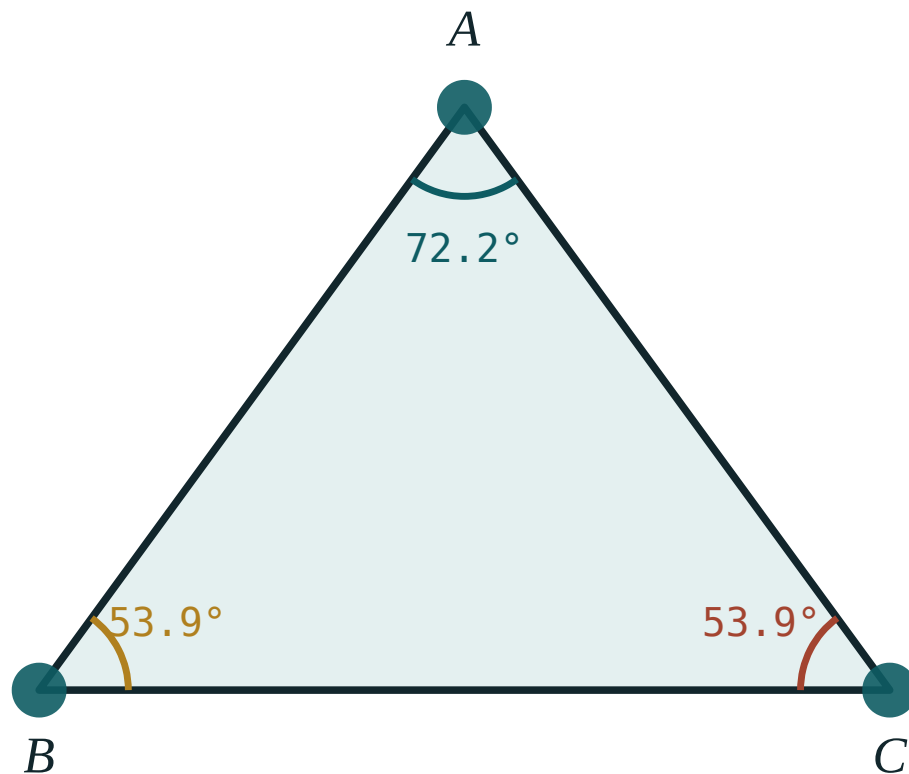
- 1 Alternate angles give $\angle ABO = \angle DCO$ and $\angle BAO = \angle CDO$.
- 2 With $AB = CD$, the second criterion gives $\triangle ABO \equiv \triangle DCO$.
- 3 CPCT: $AO = DO$ and $BO = CO$.
- 4 So each diagonal is bisected.

ANSWER

Proved

03 Interactive model

Measure the width of the corridor from inside the classroom using a tape and a protractor; the construction on paper gives an answer that can then be checked directly.

A side and two angles $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$ **04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Second criterion	Ikkinchi alomat	Второй признак
Two angles and a side	Ikki burchak va tomon	Два угла и сторона
Adjacent side	Yondosh tomon	Прилежащая сторона
Angle sum	Burchaklar yig'indisi	Сумма углов
Surveying	Yer o'lchash	Съёмка местности
Inaccessible	Yetib bo'lmaydigan	Недоступный
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The second criterion needs:

three sides

a side and two angles

two sides and an angle

three angles

The third angle:

must be given

follows from the sum

is irrelevant

is always 90°

Must the side lie between the angles?

yes

no

only sometimes

it depends on the size

Three angles alone give:

congruence

similarity

nothing

equal areas

Parallel lines supply:

equal sides

equal angles

right angles

nothing

Surveying a river width uses:

the first criterion

the second criterion

the triangle inequality

Pythagoras

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The second criterion in letters

ANSWER ASA

2. A side of 7 with 50° and 60° on it: third angle

ANSWER 70°

3. Two triangles with that data

ANSWER Congruent

4. Three equal angles give

ANSWER Similarity only

5. Alternate angles come from

ANSWER Parallel lines

6. Angles 50° , 70° : the third

ANSWER 60°

7. Is a side always needed?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. A side of 7, 50° on it and 70° opposite: congruent?

ANSWER Yes

2. $AB \parallel CD$, $AB = CD$: which criterion?

ANSWER The second

3. What follows about AO and DO ?

ANSWER They are equal

4. Two triangles with $\angle A = \angle D$, $\angle B = \angle E$, $AB = DE$

ANSWER Congruent

5. Two triangles with $\angle A = \angle D$, $\angle B = \angle E$ only

ANSWER Similar only

6. Angles 40° , 40° with the included side 9

ANSWER Congruent to any such triangle

7. The third angle there

ANSWER 100°

Hard · stretch and reason

7 PROBLEMS

1. Prove that the diagonals of a parallelogram bisect each other

ANSWER ASA with alternate angles

2. Prove that a point on the bisector of an angle is equidistant from the arms

ANSWER ASA, then CPCT

3. Prove that the base angles of an isosceles triangle are equal, using ASA

ANSWER Compare the triangle with itself, reversed

4. A triangle with $\angle A = 3x$, $\angle B = 2x$, $\angle C = x$

ANSWER 90° , 60° , 30°

5. Why is ASA enough but SSA not?

ANSWER The third angle is forced; the third side is not

6. Two triangles share a side and have two pairs of equal angles

ANSWER Congruent

7. How many measurements are needed to fix a triangle?

ANSWER Three, including at least one length

07 Homework

Homework — 5 tasks

Mark the equal angles and the equal side on the figure before writing the proof.

- Two triangles have a side of 9 with angles 45° and 65° on it. Are they congruent?
- Find the third angle in that triangle.
- Prove that the diagonals of a parallelogram bisect each other.
- Explain why three angles alone do not prove congruence.
- Describe how to measure the width of a river using the second criterion.

The third criterion of congruence of triangles

Three sides and nothing else — the criterion that makes a triangle rigid.

LESSON 53–54

2 × 40 MIN

MATEMATIKA 7, §23

S8 5

LESSON CLOCK

13'

23'

22'

18'

4'

The criterion	13 min
Rigidity	23 min
Using it	22 min
Choosing a criterion	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and apply the SSS criterion.
- Use it to prove properties of the rhombus and the kite.
- Explain why a triangle is rigid and a quadrilateral is not.
- Choose between the three criteria from the given data.

TEXTBOOK

National	pp. 143–148
Cambridge	Stage 8, pp. 48–56
Workbook	Exercise 5.1

01

Explanation

The criterion

If the three sides of one triangle are equal to the three sides of another, the triangles are congruent

$$AB = A_1B_1, BC = B_1C_1, CA = C_1A_1 \implies \triangle ABC \equiv \triangle A_1B_1C_1$$

No angle is mentioned at all, and yet all three angles turn out equal. Three lengths determine a triangle completely.

THIS IS THE CRITERION A COMPUTER WOULD USE

It needs no reasoning about the figure — only three comparisons. Every question that supplies six lengths and no angles is answered by it.

Rigidity

A framework of three rods joined at their ends cannot change shape; one of four rods can.

FRAMEWORK	RIGID?	WHY
triangle	yes	three sides fix it — the third criterion
quadrilateral	no	it can be pushed into a parallelogram
quadrilateral with one diagonal	yes	it becomes two triangles

This is why bridges, roof trusses and pylons are built from triangles, and why a garden gate is braced with a diagonal.

THE THIRD CRITERION IS AN ENGINEERING FACT

Every triangulated structure in the world depends on it. The theorem and the brace on a gate are the same statement.

Using it

Problem. In a rhombus $ABCD$, prove that the diagonal AC bisects the angles at A and C .

STATEMENT	REASON
$AB = AD$	a rhombus has four equal sides
$CB = CD$	the same
$AC = AC$	common
$\triangle ABC \equiv \triangle ADC$	third criterion
$\angle BAC = \angle DAC$	CPCT

THE COMMON SIDE AGAIN

As in the first criterion, the shared side supplies the third equality without being given. Look for it first in every figure.

Choosing a criterion

GIVEN	CRITERION
two sides and the angle between them	the first (SAS)
a side and two angles	the second (ASA)
three sides	the third (SSS)
two sides and a non-included angle	none — not enough
three angles	none — similarity only

NAME THE CRITERION IN EVERY PROOF

A proof that reaches the right conclusion without naming which criterion was used is incomplete, and is marked as such in both the national and the Cambridge papers.

02

Worked examples

EXAMPLE 1 Two triangles have sides 5, 6, 7 and 7, 5, 6. Are they congruent?

1

Sort both lists: 5, 6, 7 and 5, 6, 7.

2

All three pairs are equal.

3

By the third criterion, congruent.

4

The order in which they were written does not matter.

ANSWER

Yes, by SSS

EXAMPLE 2 In rhombus $ABCD$, prove $\triangle ABC \equiv \triangle ADC$.

- ① $AB = AD$ — all sides of a rhombus are equal.
- ② $CB = CD$ — the same.
- ③ AC is common.
- ④ Third criterion.

ANSWER

Proved by SSS

EXAMPLE 3 Why is a four-rod framework not rigid, and how is it braced?

- ① Four sides do not determine the angles.
- ② The frame can be pushed out of square.
- ③ Adding one diagonal makes two triangles.
- ④ Each is rigid by SSS, so the whole frame is.

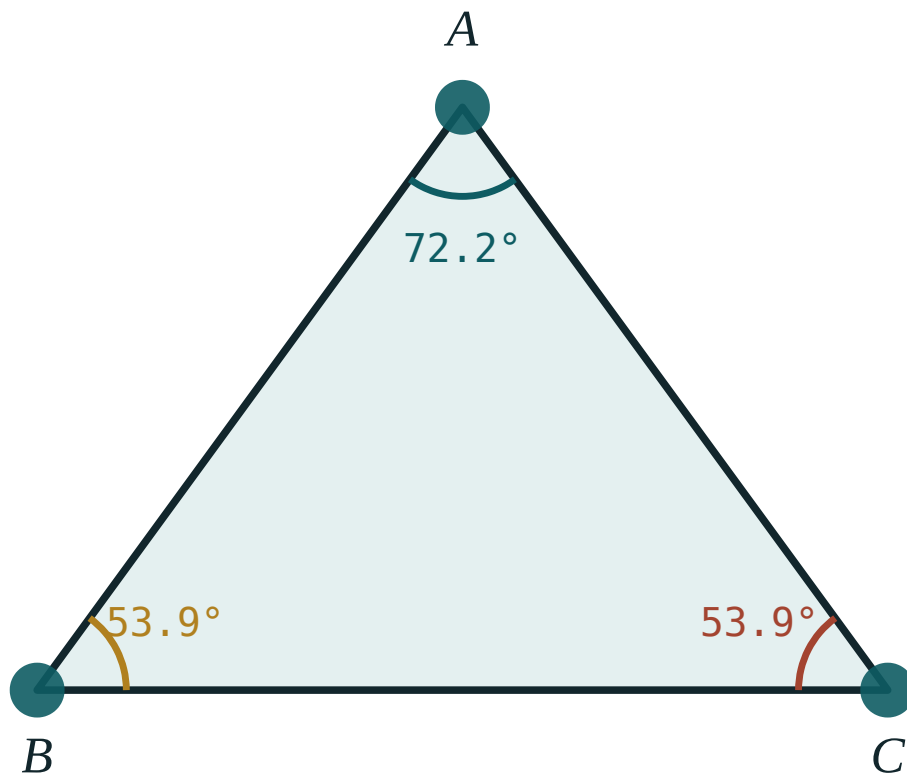
ANSWER

Add a diagonal

03 Interactive model

Pin three strips of card into a triangle and four into a quadrilateral; the class pushes both, and the third criterion becomes visible in ten seconds.

Three sides fix the triangle

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Third criterion	Uchinchi alomat	Третий признак
Three sides	Uchta tomon	Три стороны
Rigid	Qattiq	Жёсткий
Rhombus	Romb	Ромб
Kite	Deltoid	Дельтоид
Diagonal	Diagonal	Диагональ
Framework	Karkas	Каркас
To choose	Tanlash	Выбрать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The third criterion needs:

three angles

three sides

two sides and an angle

a side and two angles

A triangular framework is:

flexible

rigid

unstable

impossible

A quadrilateral framework is made rigid by:

a fifth rod along a side

a diagonal

nothing

a shorter rod

Sides 5, 6, 7 and 7, 5, 6 give:

congruence

similarity only

nothing

an error

Two sides and a non-included angle give:

congruence

not necessarily

similarity

nothing at all

A complete proof must:

give the answer

name the criterion

include measurements

be short

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The third criterion in letters

ANSWER SSS

2. Sides 5, 6, 7 and 5, 6, 7

ANSWER Congruent

3. Sides 5, 6, 7 and 5, 6, 8

ANSWER Not congruent

4. Is a triangle rigid?

ANSWER Yes

5. Is a quadrilateral rigid?

ANSWER No

6. How is it braced?

ANSWER With a diagonal

7. A shared side gives

ANSWER An equality for free

Medium · the standard question

7 PROBLEMS

1. Sides 5, 6, 7 and 7, 5, 6

ANSWER Congruent

2. Which criterion for six lengths?

ANSWER The third

3. Which for a side and two angles?

ANSWER The second

4. Which for two sides and the included angle?

ANSWER The first

5. Which for three angles?

ANSWER None

6. In a rhombus, $\triangle ABC$ and $\triangle ADC$

ANSWER Congruent by SSS

7. What follows about $\angle BAC$ and $\angle DAC$?

ANSWER They are equal

Hard · stretch and reason

7 PROBLEMS

1. Prove that a diagonal of a rhombus bisects its angles

ANSWER SSS, then CPCT

2. Prove that the diagonals of a rhombus are perpendicular

ANSWER Congruent triangles at the centre

3. Prove that a kite has one diagonal as an axis of symmetry

ANSWER SSS on the two triangles

4. Why is a bridge built of triangles?

ANSWER Only triangles are rigid

5. Two triangles with sides a, b, c and c, b, a

ANSWER Congruent

6. Can two triangles have five equal elements and not be congruent?

ANSWER Yes — similar triangles can

7. How many rods brace a quadrilateral?

ANSWER One diagonal

07 Homework

Homework — 5 tasks

Sort both lists of sides before comparing, and name the criterion in every proof.

- Are triangles with sides 8, 9, 10 and 10, 8, 9 congruent?
- Prove that a diagonal of a rhombus bisects two of its angles.
- Explain why a triangle is rigid and a quadrilateral is not.
- State which criterion applies to each of the five cases in the table.
- Prove that the diagonals of a rhombus are perpendicular.

Practical exercises

— constructions

with compasses

and a straightedge

A Cambridge insert: the four constructions every later proof depends on.

LESSON 55

1 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 5.3

LESSON CLOCK

8'

14'

12'

6'

The tools	8 min
The four constructions	14 min
Building a triangle	12 min
Homework	6 min

LEARNING OBJECTIVES

- Copy a segment and an angle with compasses.
- Construct the perpendicular bisector of a segment.
- Construct the bisector of an angle.
- Construct a triangle from three given sides.

TEXTBOOK

National	pp. 149–151
Cambridge	Stage 8, pp. 53–56
Workbook	Exercise 5.3

01

Explanation

The tools

A **construction** uses only two tools: compasses to draw circles and to carry lengths, and a straightedge to draw lines. The straightedge is **not** a ruler — no measuring is allowed.

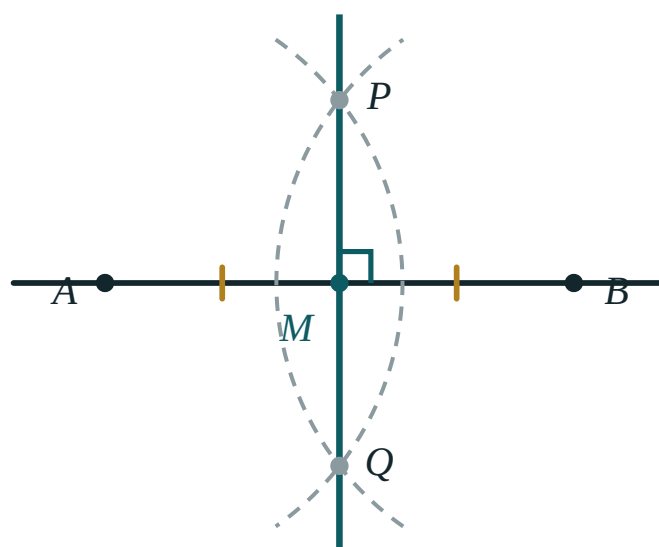
TOOL	MAY BE USED TO	MAY NOT
compasses	draw an arc, carry a length	measure in centimetres
straightedge	join two points	measure

LEAVE EVERY ARC VISIBLE

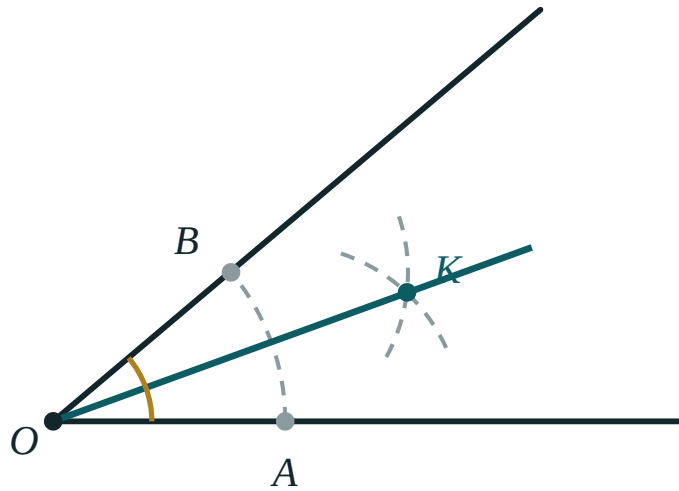
The arcs are the working. A finished construction with the arcs rubbed out cannot be marked, and scores less than an untidy one that shows them.

The four constructions

CONSTRUCTION	METHOD
copy a segment	open the compasses to it and step it off on a ray
copy an angle	one arc across both arms of the original and of the copy, then transfer the chord
perpendicular bisector of AB	equal arcs from A and from B , both sides; join the crossings
bisector of an angle	one arc across both arms, then equal arcs from the two points; join to the vertex



Equal arcs from both ends give the perpendicular bisector.



One arc across the arms, then two equal arcs, gives the bisector.

ALL FOUR ARE THE SAME IDEA

Two points equidistant from the same pair of points determine a line of symmetry. Every one of the four constructions is that statement applied somewhere.

Building a triangle

Given three sides a , b , c :

1. draw a segment $BC = a$;
2. from B , an arc of radius c ;
3. from C , an arc of radius b ;
4. the crossing is A ; join AB and AC .

By the third criterion, every triangle built this way from the same three lengths is congruent to every other — so the construction gives **the** triangle, not **a** triangle.

THE ARCS MUST ACTUALLY CROSS

If the two short sides together are less than the long one, the arcs miss and no triangle exists. The construction fails exactly when the triangle inequality fails.

02 Worked examples

EXAMPLE 1 Describe the construction of the perpendicular bisector of a segment AB .

- 1 Open the compasses to more than half of AB .
- 2 Draw arcs from A above and below the segment.
- 3 Draw the same arcs from B .
Same radius.
- 4 Join the two crossings — that line is the perpendicular bisector.

ANSWER

Equal arcs from both ends, joined

EXAMPLE 2 Construct a triangle with sides 4 cm , 5 cm and 6 cm .

- 1 Draw $BC = 6\text{ cm}$.
The longest side is easiest.
- 2 Arc of 4 cm from B .
- 3 Arc of 5 cm from C .
- 4 Join the crossing to both ends.
The arcs cross since $4 + 5 > 6$.

ANSWER

A unique triangle

EXAMPLE 3 Why does no triangle exist with sides 3, 4, 9?

- ① Draw the 9 *cm* side.
- ② Arcs of 3 and 4 from its ends.
- ③ $3 + 4 = 7 < 9$ — the arcs fall short.
- ④ They never meet, so there is no third vertex.

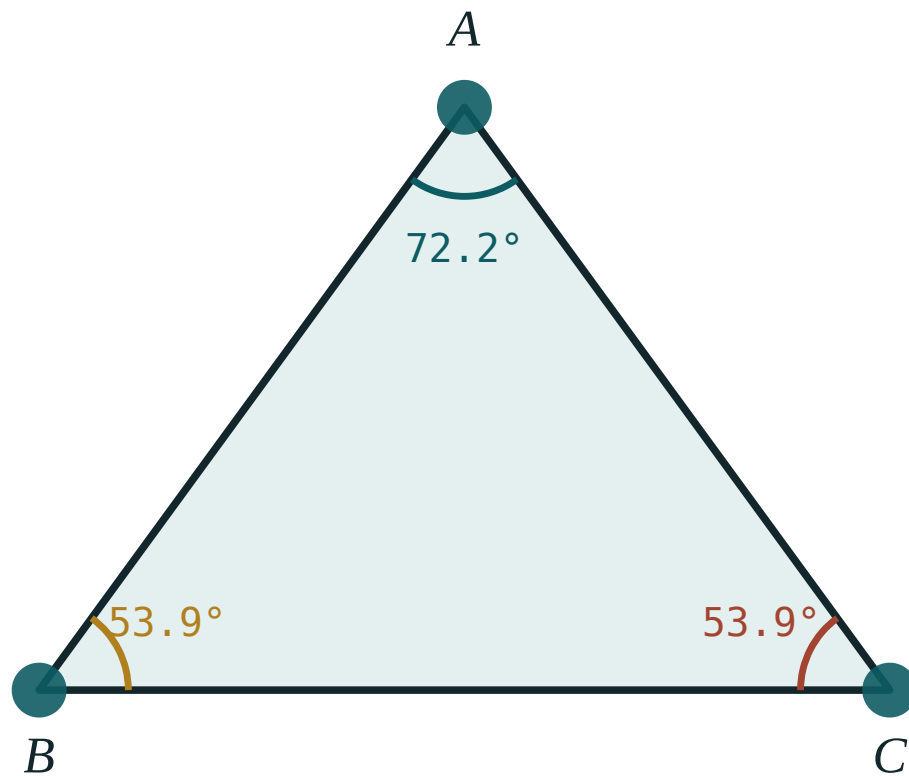
ANSWER

The arcs do not cross

03 Interactive model

Have the class construct the same triangle from the same three lengths and then lay the sheets on top of one another; every triangle coincides, and the third criterion is confirmed.

Building from three sides

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side **BC**largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Construction	Yasash	Построение
Compasses	Sirkul	Циркуль
Straightedge	Chizg'ich	Линейка
Arc	Yoy	Дуга
To copy	Ko'chirish	Перенести
Perpendicular bisector	O'rta perpendikulyar	Серединный перпендикуляр
Bisector	Bissektrisa	Биссектриса
Equidistant	Teng uzoqlikda	Равноудалённый

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A construction uses:

a ruler and a protractor

compasses and a straightedge

a calculator

a set square only

Construction arcs should be:

rubbed out

left visible

drawn in pen

omitted

The perpendicular bisector is found by:

measuring the midpoint

equal arcs from both ends

a protractor

folding only

A triangle from three sides is:

not always possible

always possible

possible in two ways

never unique

Sides 3, 4, 9:

one triangle

two triangles

no triangle

infinitely many

All triangles built from the same three sides are:

similar only

congruent

different

right-angled

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Tools allowed

ANSWER Compasses and a straightedge

2. May the straightedge measure?

ANSWER No

3. Should arcs be left?

ANSWER Yes

4. Sides 4, 5, 6: constructible?

ANSWER Yes

5. Sides 3, 4, 9: constructible?

ANSWER No

6. Sides 5, 5, 5: what triangle?

ANSWER Equilateral

7. Sides 3, 4, 5: what triangle?

ANSWER Right-angled

Medium · the standard question

7 PROBLEMS

1. Which construction gives the midpoint?

ANSWER The perpendicular bisector

2. Which gives two equal angles?

ANSWER The angle bisector

3. How many arcs for a perpendicular bisector?

ANSWER Four — two from each end

4. Construct 60° : method

ANSWER An equilateral triangle

5. Construct 30° : method

ANSWER Bisect 60°

6. Construct 90° : method

ANSWER A perpendicular

7. Construct 45° : method

ANSWER Bisect 90°

Hard · stretch and reason

7 PROBLEMS

1. Construct 15°

ANSWER Bisect 30°

2. Construct 75°

ANSWER $60^\circ + 15^\circ$

3. Construct an isosceles triangle with base 5 and lateral side 7

ANSWER Two arcs of 7

4. Why must the compasses open beyond half of AB ?

ANSWER Otherwise the arcs do not meet

5. Construct the centre of a given circle

ANSWER Two chords and their perpendicular bisectors

6. Construct a triangle from two sides and the included angle

ANSWER Copy the angle, step off the sides

7. Which construction cannot be done with these tools?

ANSWER Trisecting a general angle

07 Homework

Homework — 5 tasks

Leave every arc on the page; the arcs are the answer.

1. Construct the perpendicular bisector of a segment of 8 cm .
2. Construct the bisector of an angle of 80° .
3. Construct a triangle with sides 5 cm , 6 cm and 7 cm .
4. Construct an angle of 30° using only compasses and a straightedge.
5. Explain why no triangle can be constructed with sides 2, 3, 8.

Control work 4, and work on the mistakes

The congruence of triangles in one paper, and the geometry chapter closed.

LESSON 56–57

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 4

S8 5 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Choose the right criterion from the given data.
- Write a proof in statement-and-reason form under time.
- Use the properties of isosceles triangles and the triangle inequality.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 118–151
Cambridge	Stage 8, pp. 48–56
Workbook	Control paper M4

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Can a triangle have sides 5, 8, 14? And 6, 9, 13?	4	L46
2	An isosceles triangle has apex 44°: find the base angles	4	L49–50
3	$AB = AD$, $\angle BAC = \angle DAC$: prove $\triangle ABC \equiv \triangle ADC$	6	L47–48
4	Name the criterion for: three sides; a side and two angles; two sides and the angle between	3	L47–54
5	In a rhombus, prove that a diagonal bisects two angles	5	L53–54
6	Describe the construction of an angle bisector	3	L55

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for checking the longest side only; Q3 two for the common side and the named criterion; Q5 two for stating that all four sides of a rhombus are equal; Q6 one for “leave the arcs”.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
only two sides checked	$5 + 14 > 8 \checkmark$	check the longest: $5 + 8 < 14$
apex used as a base angle	$44^\circ, 44^\circ, 92^\circ$	$44^\circ, 68^\circ, 68^\circ$
common side omitted	two facts only	$AC = AC$
criterion not named	“so they are congruent”	“by the first criterion”
rhombus sides assumed unequal	“given $AB = AD$ ”	state that all four are equal
reasons missing	a list of true statements	a reason beside each
arcs rubbed out	a clean bisector	leave the construction visible

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter IV as one map

BLOCK	THE SENTENCE
triangles	classified by sides and by angles, independently
the triangle inequality	the longest side is less than the sum of the other two
the first criterion	two sides and the angle between them
isosceles triangles	equal sides $\iff$ equal base angles
the second criterion	a side and two angles
the third criterion	three sides — and rigidity
constructions	compasses and a straightedge, arcs left visible

LOOKING FORWARD

The rest of Quarter II is algebra: expressions, powers, monomials and polynomials. Geometry returns in Quarter III with parallel lines and the angle sum of a triangle.

02 Worked examples

EXAMPLE 1 Model answer, Q1: sides 5, 8, 14 and 6, 9, 13.

- ① Longest of the first is 14.
- ② $5 + 8 = 13 < 14$ — no triangle.
- ③ Longest of the second is 13.
- ④ $6 + 9 = 15 > 13$ — yes.

ANSWER

No; yes

EXAMPLE 2 Model answer, Q3: $AB = AD$, $\angle BAC = \angle DAC$.

- ① $AB = AD$ — given.
- ② $\angle BAC = \angle DAC$ — given, and it is the included angle.
- ③ $AC = AC$ — common side.
The mark most often missed.
- ④ $\triangle ABC \equiv \triangle ADC$ by the first criterion.

ANSWER

Proved by SAS

EXAMPLE 3 Model answer, Q5: a diagonal of a rhombus bisects two angles.

- ① All four sides of a rhombus are equal: $AB = AD$ and $CB = CD$.
- ② AC is common.
- ③ $\triangle ABC \equiv \triangle ADC$ by the third criterion.
- ④ CPCT gives $\angle BAC = \angle DAC$ and $\angle BCA = \angle DCA$.

ANSWER

Proved by SSS

03 Interactive model

Return Q3 with the reasons column blank and have the class fill it in; they discover that the reasons carry most of the marks.

Chapter IV in twelve questions

A triangle with two equal sides is:

scalene

isosceles

equilateral only

right

Can 2, 3, 6 be a triangle?

yes

no

sometimes

only if right

Congruent triangles have how many equal pairs?

3

4

6

2

The first criterion needs:

three sides

two sides and the included angle

a side and two angles

three angles

Base angles of an isosceles triangle are:

unequal

equal

right

obtuse

Apex 40° gives base angles:

40°

70°

100°

140°

The second criterion needs:

three sides

a side and two angles

three angles

two sides

The third criterion needs:

three sides

three angles

two sides

a side and an angle

A triangular framework is:

flexible

rigid

unstable

impossible

A common side in a figure gives:

nothing

an equality

an angle

a contradiction

Three equal angles give:

congruence

similarity only

nothing

rigidity

Construction arcs should be:

rubbed out

left visible

in pen

omitted

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Congruence	Tenglik	Равенство
Criterion	Alomat	Признак
Isosceles	Teng yonli	Равнобедренный
Triangle inequality	Uchburchak tengsizligi	Неравенство треугольника
Common side	Umumiy tomon	Общая сторона
Reason	Asos	Обоснование
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For Q1 it is enough to check:

all three sums

the longest side

the shortest side

the perimeter

Q2 needs the angles to add to:

90°

180°

270°

360°

The missing fact in Q3 is:

an angle

the common side

a third side

nothing

Q5 uses:

SAS

ASA

SSS

none

Q6 must mention:

a protractor

leaving the arcs

measuring

a ruler

Work on the mistakes means:

fix the wrong line

rewrite the solution

copy the answer

skip it

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Can 5, 8, 14 be a triangle?

ANSWER No

2. Can 6, 9, 13 be a triangle?

ANSWER Yes

3. Apex 44° : base angles

ANSWER 68°

4. Three sides: which criterion?

ANSWER The third

5. A side and two angles: which?

ANSWER The second

6. Two sides and the included angle: which?

ANSWER The first

7. Three angles: which?

ANSWER None

Medium · the standard question

7 PROBLEMS

1. $AB = AD$, $\angle BAC = \angle DAC$: the third fact

ANSWER AC common

2. That proof uses

ANSWER The first criterion

3. A rhombus diagonal proof uses

ANSWER The third criterion

4. Base angle 68° : apex

ANSWER 44°

5. An isosceles triangle with perimeter 28 and base 8

ANSWER Sides 10, 10, 8

6. Sides 7, 7, 7: angles

ANSWER 60° each

7. Which framework is rigid?

ANSWER The triangle

Hard · stretch and reason

7 PROBLEMS

1. Range of c with $a = 6$, $b = 11$

ANSWER $5 < c < 17$

2. Integer values of c there

ANSWER 6 to 16

3. An isosceles triangle with an exterior base angle of 115° : the apex

ANSWER 50°

4. Prove the base angles equal, using the bisector from the apex

ANSWER SAS then CPCT

5. Prove the diagonals of a rhombus are perpendicular

ANSWER SSS then equal adjacent angles

6. Two triangles with $AB = DE$, $\angle A = \angle D$, $\angle B = \angle E$

ANSWER Congruent by ASA

7. Two triangles with $AB = DE$, $BC = EF$, $\angle A = \angle D$

ANSWER Not necessarily congruent

07 Homework

Homework — 5 tasks

Rewrite in full every proof that lost a mark, with a reason beside every statement.

- Can a triangle have sides 4, 7, 12? And 5, 7, 11?
- An isosceles triangle has apex angle 52° . Find the base angles.
- Prove that a diagonal of a rhombus bisects two of its angles.
- Name the criterion that applies to each of the three standard sets of data.
- Describe the construction of the bisector of an angle.

Algebraic expressions

Letters standing for numbers — and the vocabulary the rest of the year needs.

LESSON 58

1 × 40 MIN

MATEMATIKA 7, §24

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

Words into symbols	10 min
The parts of an expression	12 min
Evaluating	12 min
Homework	6 min

LEARNING OBJECTIVES

- Write a worded statement as an algebraic expression.
- Name the terms, factors and coefficients of an expression.
- Evaluate an expression for given values of the letters.
- Distinguish an expression from an equation.

TEXTBOOK

National	pp. 152–155
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

Words into symbols

WORDS	EXPRESSION
five more than x	$x + 5$
five less than x	$x - 5$
five times x	$5x$
x divided by five	$\frac{x}{5}$
the square of x , less three	$x^2 - 3$
twice the sum of x and y	$2(x + y)$

“TWICE THE SUM” NEEDS BRACKETS

$2(x + y)$ and $2x + y$ are different expressions. The bracket is what makes the doubling apply to the whole sum.

The parts of an expression

An expression is built from **terms** separated by + and −. Each term is a product of a number and some letters.

EXPRESSION	TERMS	COEFFICIENTS
$3x + 5y - 7$	$3x, 5y, -7$	$3, 5$
$-4a^2 + a$	$-4a^2, a$	$-4, 1$
$x - \frac{y}{2}$	$x, -\frac{y}{2}$	$1, -\frac{1}{2}$

A term with no letter is a **constant**. A coefficient of 1 or −1 is not written: a means $1a$ and $-a$ means $-1a$.

THE SIGN BELONGS TO THE TERM

In $3x - 7y$ the second term is $-7y$, with coefficient -7 . Reading the coefficient as 7 loses the sign in every later calculation.

Evaluating

To **evaluate** an expression, substitute the given numbers and compute, respecting the order of operations.

EXPRESSION	$X = 3, Y = -2$	VALUE
$2x + y$	$6 + (-2)$	4
$x^2 - y$	$9 - (-2)$	11
xy	$3 \times (-2)$	-6
$2(x + y)$	$2(1)$	2

PUT NEGATIVE VALUES IN BRACKETS

Writing $x^2 - y$ as $9 - -2$ invites an error; $9 - (-2)$ does not. The brackets cost nothing and prevent the commonest slip in substitution.

02 Worked examples**EXAMPLE 1** Write as expressions: seven more than twice x ; the square of the sum of a and b .

- ① Twice x is $2x$.
- ② Seven more: $2x + 7$.
- ③ The sum is $a + b$.
- ④ Its square: $(a + b)^2$.
Brackets are essential.

ANSWER $2x + 7$ and $(a + b)^2$ **EXAMPLE 2** Name the terms and coefficients of $3x + 5y - 7$.

- ① Terms are separated by $+$ and $-$.
- ② $3x$, $5y$, -7 .
- ③ Coefficients: 3 and 5.
- ④ -7 is a constant term.

ANSWER

Three terms; coefficients 3 and 5

EXAMPLE 3 Evaluate $x^2 - y$ and $2(x + y)$ at $x = 3, y = -2$.

$$1 \quad 3^2 - (-2) = 9 + 2$$

Brackets round the negative.

$$2 \quad = 11$$

$$3 \quad 2(3 + (-2)) = 2(1)$$

$$4 \quad = 2$$

ANSWER

11 and 2

03 Interactive model

Read six phrases aloud and have the class write the expressions before any of them is discussed; the errors that appear are exactly the ones to teach.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Algebraic expression	Algebraik ifoda	Алгебраическое выражение
Variable	O'zgaruvchi	Переменная
Term	Had	Член
Coefficient	Koeffitsiyent	Коэффициент
Constant	O'zgarmas	Постоянная
To evaluate	Qiymatini topmoq	Вычислить значение
To substitute	O'rniga qo'yimoq	Подставить
Equation	Tenglama	Уравнение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

“Five less than x ” is:

“Twice the sum of x and y ” is:

How many terms in $3x + 5y - 7$?

The coefficient of y in $3x - 7y$:

7

-7

3

1

$x^2 - y$ at $x = 3, y = -2$:

7

11

-11

5

A term with no letter is:

a coefficient

a constant

a variable

an equation

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Five more than x

ANSWER $x + 5$

2. Five less than x

ANSWER $x - 5$

3. Five times x

ANSWER $5x$

4. x divided by five

ANSWER $\frac{x}{5}$

5. Terms in $3x + 5y - 7$

ANSWER 3

6. Coefficient of x in $3x$

ANSWER 3

7. Coefficient of a in $-a$

ANSWER -1

Medium · the standard question

7 PROBLEMS

1. Twice the sum of x and y

ANSWER $2(x + y)$

2. Seven more than twice x

ANSWER $2x + 7$

3. The square of the sum of a and b

ANSWER $(a + b)^2$

4. Coefficient of y in $3x - 7y$

ANSWER -7

5. $2x + y$ at $x = 3, y = -2$

ANSWER 4

6. $x^2 - y$ at the same

ANSWER 11

7. xy at the same

ANSWER -6

Hard · stretch and reason

7 PROBLEMS

1. $x^2 + 2xy + y^2$ at $x = 3, y = -2$

ANSWER 1

2. $\frac{x+y}{x-y}$ at $x = 3, y = -2$

ANSWER $\frac{1}{5}$

3. $-a^2$ at $a = -4$

ANSWER -16

4. $(-a)^2$ at $a = -4$

ANSWER 16

5. Write: the sum of the squares of a and b

ANSWER $a^2 + b^2$

6. Write: the square of the sum of a and b

ANSWER $(a + b)^2$

7. Are they the same?

ANSWER No — try $a = b = 1$

07 Homework

Homework — 5 tasks

Put every substituted negative value in brackets.

1. Write as expressions: four less than three times x ; half the sum of a and b .
2. Name the terms and coefficients of $5a - 2b + 9$.
3. Evaluate $3x - 2y$ at $x = -1, y = 4$.
4. Evaluate $(x + y)^2$ at the same values.

5. Explain the difference between $a^2 + b^2$ and $(a + b)^2$.

Rules for removing brackets, and the coefficient

A plus leaves the signs alone; a minus changes every one of them.

LESSON 59

1 × 40 MIN

MATEMATIKA 7, §25

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

Plus and minus before a bracket	10 min
A number before a bracket	12 min
The coefficient	12 min
Homework	6 min

LEARNING OBJECTIVES

- Remove brackets preceded by a plus and by a minus sign.
- Multiply a bracket by a number using the distributive law.
- Identify and use the coefficient of a term.
- Simplify an expression by removing brackets and collecting terms.

TEXTBOOK

National	pp. 156–159
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

Plus and minus before a bracket

$a + (b - c) = a + b - c$

$a - (b - c) = a - b + c$

EXPRESSION	BRACKETS REMOVED
$5 + (x - 3)$	$5 + x - 3$
$5 - (x - 3)$	$5 - x + 3$
$-(2a - b + 4)$	$-2a + b - 4$
$7 - (-x + 2)$	$7 + x - 2$

A MINUS CHANGES EVERY SIGN INSIDE

$-(2a - b + 4) = -2a + b - 4$ — all three. Changing only the first is the error this lesson exists to prevent, and it recurs all year.

A number before a bracket

$$k(a + b) = ka + kb$$

Every term inside is multiplied — including the last one, and including the signs.

EXPRESSION	EXPANDED
$3(x + 4)$	$3x + 12$
$3(x - 4)$	$3x - 12$
$-3(x - 4)$	$-3x + 12$
$-2(3a - b + 5)$	$-6a + 2b - 10$

$- (...)$ IS $-1(...)$

The two rules of this lesson are one rule: a minus in front is multiplication by -1 , and the distributive law does the rest.

The coefficient

The **coefficient** is the numerical factor of a term, taken with its sign.

TERM	COEFFICIENT	LETTERS
$7ab$	7	ab
$-3x^2y$	-3	x^2y
a	1	a
$-b$	-1	b
$\frac{2x}{3}$	$\frac{2}{3}$	x

Terms with the **same letters to the same powers** are **like terms**, and only like terms can be collected: $3x + 5x = 8x$, but $3x + 5y$ cannot be simplified.

$3x$ AND $3x^2$ ARE NOT LIKE TERMS

The powers must match too. Adding them gives $3x + 3x^2$, which is already as simple as it gets.

02 Worked examples

EXAMPLE 1 Remove the brackets: $5 - (x - 3)$ and $-(2a - b + 4)$.

① A minus before the bracket changes every sign.

② $5 - x + 3 = 8 - x$

③ $-2a + b - 4$
All three signs.

④ Nothing further to collect in the second.

ANSWER

$8 - x$ and $-2a + b - 4$

EXAMPLE 2 Expand $-2(3a - b + 5)$.

① Multiply every term by -2 .

② $-2 \times 3a = -6a$

③ $-2 \times (-b) = +2b$

④ $-2 \times 5 = -10$, giving $-6a + 2b - 10$.

ANSWER

$-6a + 2b - 10$

EXAMPLE 3 Simplify $4(x + 2) - 3(x - 1)$.

$$① \quad 4x + 8$$

$$② \quad -3x + 3$$

Both signs change.

$$③ \quad 4x + 8 - 3x + 3$$

$$④ \quad = x + 11$$

ANSWER

$$x + 11$$

03 Interactive model

Write $-(a - b)$ on the board and ask for its value at $a = 5$, $b = 2$ before removing the brackets; the numerical check settles the sign rule.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Bracket	Qavs	Скобка
To remove brackets	Qavslarni ochish	Раскрыть скобки
Distributive law	Taqsimot qonuni	Распределительный закон
Coefficient	Koeffitsiyent	Коэффициент
Like terms	O'xshash hadlar	Подобные члены
To simplify	Soddalashtirish	Упростить
Sign change	Ishora o'zgarishi	Изменение знака
Nested brackets	Ichma-ich qavslar	Вложенные скобки

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$5 - (x - 3)$ equals:

$-(2a - b + 4)$ equals:

$-3(x - 4)$ equals:

The coefficient of $-3x^2y$:

3

-3

x^2y

1

The coefficient of $-b$:

0

1

-1

b

Are $3x$ and $3x^2$ like terms?

yes

no

sometimes

only if $x = 1$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $5 + (x - 3)$

ANSWER $x + 2$

2. $5 - (x - 3)$

ANSWER $8 - x$

3. $3(x + 4)$

ANSWER $3x + 12$

4. $3(x - 4)$

ANSWER $3x - 12$

5. $-3(x - 4)$

ANSWER $-3x + 12$

6. Coefficient of $7ab$

ANSWER 7

7. Coefficient of $-b$

ANSWER -1 **Medium · the standard question**

7 PROBLEMS

1. $-(2a - b + 4)$

ANSWER $-2a + b - 4$

2. $-2(3a - b + 5)$

ANSWER $-6a + 2b - 10$

3. $7 - (-x + 2)$

ANSWER $x + 5$

4. $4(x + 2) - 3(x - 1)$

ANSWER $x + 11$

5. $2(a - 3) + 5(a + 1)$

ANSWER $7a - 1$

6. Collect $3x + 5x - 2x$

ANSWER $6x$

7. Collect $3x + 5y$

ANSWER Already simplest

Hard · stretch and reason

7 PROBLEMS

1. $3(2x - 1) - 2(3x - 4)$

ANSWER 5

2. $-[2 - (3 - x)]$

ANSWER $1 - x$

3. $5 - 2(x - 3(x + 1))$

ANSWER $4x + 11$

4. $a - (b - (c - d))$

ANSWER $a - b + c - d$

5. Simplify $4(x + y) - 4(x - y)$

ANSWER $8y$

6. Evaluate $-(a - b)$ at $a = 5, b = 2$

ANSWER -3

7. Coefficient of $\frac{3x}{4}$

ANSWER $\frac{3}{4}$ **07 Homework****Homework — 5 tasks**

Check every expansion by substituting one easy pair of values.

1. Remove the brackets: $8 - (y - 5)$ and $-(3m - 2n + 1)$.
2. Expand $-4(2a - 3b + 1)$.
3. Simplify $5(x - 2) - 2(x + 3)$.
4. Give the coefficients in $-5a + b - \frac{c}{2}$.

5. Simplify $3(a + b) - 3(a - b)$.

Properties of the arithmetic operations

The five laws that make algebra possible, written for letters instead of numbers.

LESSON 60

1 × 40 MIN

MATEMATIKA 7, §26

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

The five laws	10 min
What is not a law	12 min
Using them	12 min
Homework	6 min

LEARNING OBJECTIVES

- State the commutative, associative and distributive laws.
- Use them to reorder and regroup a calculation.
- Recognise that subtraction and division are not commutative.
- Use the laws to simplify an expression quickly.

TEXTBOOK

National	pp. 160–163
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

The five laws

LAW	STATEMENT	SAYS
commutative (+)	$a + b = b + a$	order does not matter
commutative (×)	$ab = ba$	order does not matter
associative (+)	$(a + b) + c = a + (b + c)$	grouping does not matter
associative (×)	$(ab)c = a(bc)$	grouping does not matter
distributive	$a(b + c) = ab + ac$	the two operations connect

Two more, about the special numbers 0 and 1:

$a + 0 = a$ $a \times 1 = a$ $a \times 0 = 0$

THE DISTRIBUTIVE LAW IS THE ONLY ONE THAT MIXES + AND ×

Everything in algebra that turns a sum into a product, or a product into a sum, is that one law — expanding brackets, factorising, and collecting like terms alike.

What is not a law

CLAIM	TRUE?	COUNTER-EXAMPLE
$a - b = b - a$	no	$5 - 3 \neq 3 - 5$
$a \div b = b \div a$	no	$6 \div 2 \neq 2 \div 6$
$(a - b) - c = a - (b - c)$	no	$(9 - 4) - 2 \neq 9 - (4 - 2)$
$a + (b \times c) = (a + b)(a + c)$	no	$2 + 3 \times 4 \neq 5 \times 6$

ONE COUNTER-EXAMPLE IS A COMPLETE DISPROOF

A rule that fails once is not a rule. Testing a suspicious claim with small numbers is the fastest check in all of algebra.

Using them

The laws are used to make a calculation easier, not to make it longer.

CALCULATION	LAW USED	EASY ROUTE
$25 \times 17 \times 4$	commutative and associative	$(25 \times 4) \times 17 = 1700$
7×98	distributive	$7(100 - 2) = 686$
$36 + 47 + 64$	commutative	$(36 + 64) + 47 = 147$
$4a + 7b - a + 2b$	commutative	$3a + 9b$

LOOK FOR PAIRS THAT MAKE ROUND NUMBERS

25 with 4, 36 with 64, 98 with 100. Spotting them before starting turns a hard calculation into an easy one.

02 Worked examples

EXAMPLE 1 Compute $25 \times 17 \times 4$ in the easiest way.

① Reorder: $25 \times 4 \times 17$.

Commutative.

② Regroup: $(25 \times 4) \times 17$.

Associative.

③ $= 100 \times 17$

④ $= 1700$

ANSWER

1700

EXAMPLE 2 Compute 7×98 without a calculator.

① $98 = 100 - 2$

② $7(100 - 2)$

Distributive.

③ $= 700 - 14$

④ $= 686$

ANSWER

686

EXAMPLE 3 Show that subtraction is not associative.

- ① Test $(9 - 4) - 2$ and $9 - (4 - 2)$.
- ② $5 - 2 = 3$
- ③ $9 - 2 = 7$
- ④ $3 \neq 7$ — one counter-example is enough.

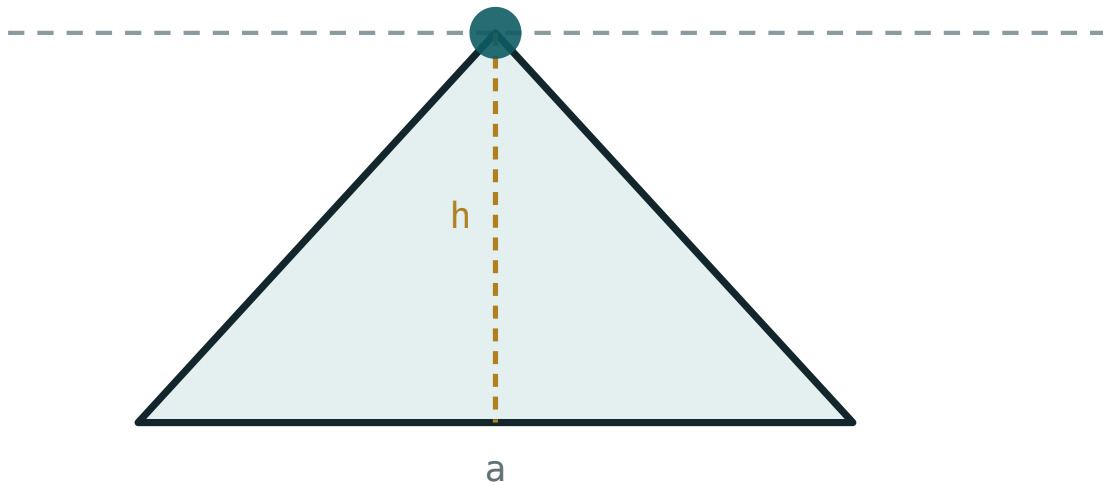
ANSWER

Disproved

03 Interactive model

Ask for $25 \times 17 \times 4$ with no calculator; the class that reorders finishes in five seconds and the rest do not.

The distributive law as an area

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Commutative law	O'rin almashtirish qonuni	Переместительный закон
Associative law	Guruhlash qonuni	Сочетательный закон
Distributive law	Taqsimot qonuni	Распределительный закон
Identity element	Neytral element	Нейтральный элемент
To reorder	Tartibini o'zgartirish	Переставить
To regroup	Guruhlarni o'zgartirish	Перегруппировать
Not commutative	Kommutativ emas	Не коммутативно
Counter-example	Qarama-qarshi misol	Контрпример

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$a + b = b + a$ is:

associative

commutative

distributive

not a law

$(ab)c = a(bc)$ is:

associative

commutative

distributive

not a law

$a(b + c) = ab + ac$ is:

associative

commutative

distributive

not a law

Is subtraction commutative?

yes

no

sometimes

only for positives

The easiest route to $25 \times 17 \times 4$:

left to right

pair 25 with 4

pair 17 with 4

a calculator

One counter-example:

weakens a rule

disproves it

proves it

means nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Name the law: $a + b = b + a$

ANSWER Commutative

2. Name the law: $(ab)c = a(bc)$

ANSWER Associative

3. Name the law: $a(b + c) = ab + ac$

ANSWER Distributive

4. Is $a - b = b - a$?

ANSWER No

5. Is $a \div b = b \div a$?

ANSWER No

6. $a + 0$

ANSWER a

7. $a \times 1$

ANSWER a

Medium · the standard question

7 PROBLEMS

1. $25 \times 17 \times 4$

ANSWER 1700

2. 7×98

ANSWER 686

3. $36 + 47 + 64$

ANSWER 147

4. $8 \times 125 \times 6$

ANSWER 6000

5. 12×101

ANSWER 1212

6. Simplify $4a + 7b - a + 2b$

ANSWER $3a + 9b$

7. Simplify $5x - 2y + 3y - x$

ANSWER $4x + y$ **Hard · stretch and reason**

7 PROBLEMS

1. $99 \times 99 + 99$

ANSWER 9900

2. $37 \times 25 \times 4$

ANSWER 3700

3. $16 \times 47 + 16 \times 53$

ANSWER 1600

4. Show that $(a - b) - c \neq a - (b - c)$

ANSWER $(9-4)-2 = 3$, $9-(4-2) = 7$

5. Simplify $2(3a - b) - 3(a - 2b)$

ANSWER $3a + 4b$

6. 101×99

ANSWER 9999

7. Why is the distributive law special?

ANSWER It is the only one linking + and $\times$ **07 Homework****Homework — 5 tasks**

Look for the easy pairing before computing anything.

1. Compute $4 \times 37 \times 25$ using the laws.
2. Compute 9×102 using the distributive law.
3. Compute $58 + 79 + 42$ by reordering.
4. Show with an example that division is not commutative.
5. Simplify $6a - 2b + 3b - 4a$.

Powers with a natural exponent

Repeated multiplication, written short — and the vocabulary that goes with it.

LESSON 61–62

2 × 40 MIN

MATEMATIKA 7, §27

S8 6.1

LESSON CLOCK

13'

23'

22'

18'

4'

The definition	13 min
Signs and fractions	23 min
Order of operations	22 min
Squares and cubes	18 min
Homework	4 min

LEARNING OBJECTIVES

- Define a^n for a natural exponent and name the base and the exponent.
- Compute powers of positive, negative and fractional bases.
- Use the correct order of operations with powers.
- Recognise squares and cubes of the small numbers.

TEXTBOOK

National	pp. 164–170
Cambridge	Stage 8, pp. 58–62
Workbook	Exercise 6.1

01

Explanation

The definition

$$a^n = a \times a \times \dots \times a \text{ (} n \text{ factors), } n \in \mathbb{N}$$

a is the **base**, n the **exponent**. $a^1 = a$, and a^2 is read “ a squared”, a^3 “ a cubed”.

POWER	MEANS	VALUE
2^5	$2 \times 2 \times 2 \times 2 \times 2$	32
3^4	$3 \times 3 \times 3 \times 3$	81
10^3	$10 \times 10 \times 10$	1000
7^1	7	7

2⁵ IS NOT 2 × 5

It is 32, not 10. The exponent counts the factors; it is not a multiplier. Reading the power aloud as “two multiplied by itself five times over” prevents the confusion.

Signs and fractions

POWER	VALUE	WHY
$(-2)^3$	-8	three negative factors
$(-2)^4$	16	four negative factors
-2^4	-16	only the 2 is raised
$(\frac{2}{3})^3$	$\frac{8}{27}$	numerator and denominator both
$(0.1)^2$	0.01	0.1×0.1

$$(-a)^n = a^n \text{ for even } n, (-a)^n = -a^n \text{ for odd } n$$

THE BRACKETS DECIDE WHAT IS RAISED

$(-2)^4 = 16$ but $-2^4 = -16$. This distinction is tested every year from now to Grade 11.

Order of operations

Powers come after brackets and before multiplication.

EXPRESSION	WORKING	VALUE
3×2^3	3×8	24
$(3 \times 2)^3$	6^3	216
$2 + 3^2$	$2 + 9$	11
$(2 + 3)^2$	5^2	25
$5^2 - 3 \times 2^2$	$25 - 12$	13

A BRACKET IS THE ONLY WAY TO RAISE A SUM

$(2 + 3)^2$ is 25; $2 + 3^2$ is 11; and $2^2 + 3^2$ is 13. Three different expressions, and the brackets are what distinguish them.

Squares and cubes

N	1	2	3	4	5	6	7	8	9	10
n^2	1	4	9	16	25	36	49	64	81	100
n^3	1	8	27	64	125	216	343	512	729	1000

Worth adding: $2^5 = 32$, $2^6 = 64$, $2^7 = 128$, $2^8 = 256$, $2^{10} = 1024$, and $3^4 = 81$, $5^4 = 625$.

THESE TWO ROWS SAVE MORE TIME THAN ANY RULE

They are used in factorising, in simplifying surds, in Pythagoras and in every prime factorisation. Learning them once repays the effort for five years.

02

Worked examples

EXAMPLE 1 Compute $(-2)^4$, -2^4 and $(-2)^5$.

1

$(-2)^4$: four negative factors — positive.

2

$= 16$

3

-2^4 means $-(2^4)$ — only the 2 is raised.

4

$= -16$; and $(-2)^5 = -32$.
Five factors — odd.

ANSWER

16, -16, -32

EXAMPLE 2 Compute $5^2 - 3 \times 2^2$.

- ① Powers first: 25 and 4.
- ② Then multiplication: $3 \times 4 = 12$.
- ③ Then subtraction.
- ④ $25 - 12 = 13$

ANSWER

13

EXAMPLE 3 Compute $(\frac{2}{3})^3$ and $(0.1)^2$.

- ① Raise the numerator and the denominator.
- ② $\frac{2^3}{3^3} = \frac{8}{27}$
- ③ 0.1×0.1
- ④ $= 0.01$

ANSWER

 $\frac{8}{27}$ and 0.01**03 Interactive model**

Fold a sheet of paper in half repeatedly and count the layers; the class reaches 2^7 before the paper stops folding, and the growth is felt rather than stated.

Powers of a number

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Power	Daraja	Степень
Base	Asos	Основание
Exponent	Ko'rsatkich	Показатель
Square	Kvadrat	Квадрат
Cube	Kub	Куб
To raise to a power	Darajaga ko'tarish	Возвести в степень
Repeated multiplication	Takroriy ko'paytirish	Повторное умножение
Order of operations	Amallar tartibi	Порядок действий

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

2^5 equals:

In a^n , n is the:

$(-2)^4$ equals:

-2^4 equals:

16

-16

8

-8

$(2 + 3)^2$ equals:

11

13

25

36

$(\frac{2}{3})^3$ equals:

$\frac{2}{27}$

$\frac{8}{27}$

$\frac{6}{9}$

$\frac{8}{3}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 2^5

ANSWER 32

2. 3^4

ANSWER 81

3. 10^3

ANSWER 1000

4. 7^2

ANSWER 49

5. 5^3

ANSWER 125

6. $(-2)^3$

ANSWER -8

7. $(-2)^4$

ANSWER 16

Medium · the standard question

7 PROBLEMS

1. -2^4

ANSWER -16

2. $(\frac{2}{3})^3$

ANSWER $\frac{8}{27}$

3. $(0.1)^2$

ANSWER 0.01

4. 3×2^3

ANSWER 24

5. $(3 \times 2)^3$

ANSWER 216

6. $2 + 3^2$

ANSWER 11

7. $5^2 - 3 \times 2^2$

ANSWER 13 **Hard · stretch and reason**

7 PROBLEMS

1. $(2 + 3)^2 - 2^2 - 3^2$

ANSWER 12

2. $(-3)^2 + (-3)^3$

ANSWER -18

3. $2^8 \div 2^5$

ANSWER 8

4. $(0.2)^3$

ANSWER 0.008

5. $(-\frac{1}{2})^4$

ANSWER $\frac{1}{16}$

6. Which is larger, 2^{10} or 10^3 ?

ANSWER $2^{10} = 1024$

7. $3^2 \times 3^3$

ANSWER 243

07 Homework

Homework — 5 tasks

Write out the factors for any power you are unsure of.

1. Compute 2^6 , 3^5 and 10^4 .

2. Compute $(-3)^3$, $(-3)^4$ and -3^4 .

3. Compute $(\frac{3}{4})^2$ and $(0.3)^2$.

4. Compute $4^2 + 2 \times 3^2$.

5. Learn the squares to 12 and the cubes to 6, and write them out from memory.

Properties of powers with a natural exponent

Five laws — and every one of them is just counting the factors.

LESSON 63–64

2 × 40 MIN

MATEMATIKA 7, §28

S8 6.2

LESSON CLOCK

13'

23'

22'

18'

4'

Multiplying and dividing	13 min
Powers of powers	23 min
Products and quotients	22 min
Putting them together	18 min
Homework	4 min

LEARNING OBJECTIVES

- State and prove the laws of indices for natural exponents.
- Multiply and divide powers with the same base.
- Raise a power, a product and a quotient to a power.
- Simplify an expression containing several powers.

TEXTBOOK

National	pp. 171–177
Cambridge	Stage 8, pp. 60–65
Workbook	Exercise 6.2

01

Explanation

Multiplying and dividing

$$a^m \cdot a^n = a^{m+n} \quad a^m \div a^n = a^{m-n} \quad (m > n)$$

Why. $a^3 \cdot a^2 = (aaa)(aa) = a^5$: the exponents count the factors, and multiplying puts the two lists end to end.

EXPRESSION	WORKING	ANSWER
$2^3 \cdot 2^4$	2^{3+4}	$2^7 = 128$
$x^5 \cdot x$	x^{5+1}	x^6
$3^7 \div 3^4$	3^{7-4}	$3^3 = 27$
$a^6 \div a^6$	a^0	1

THE BASES MUST BE THE SAME

$2^3 \cdot 3^2$ cannot be combined — the laws say nothing about it. Only $2^3 \cdot 2^2$ becomes 2^5 .

Powers of powers

$$(a^m)^n = a^{mn}$$

Why. $(a^3)^2 = a^3 \cdot a^3 = a^6$: n copies of m factors is mn factors.

EXPRESSION	ANSWER
$(2^3)^2$	$2^6 = 64$
$(x^4)^3$	x^{12}
$(a^2)^5$	a^{10}

$(A^3)^2$ AND $A^3 \cdot A^2$ ARE DIFFERENT

The first is a^6 , the second a^5 . Multiply the exponents for a power of a power; add them for a product.

Products and quotients

$$(ab)^n = a^n b^n \left(\frac{a}{b} \right)^n = \frac{a^n}{b^n}$$

EXPRESSION	ANSWER
$(2x)^3$	$8x^3$
$(-3a)^2$	$9a^2$
$\left(\frac{x}{2}\right)^4$	$\frac{x^4}{16}$
$(2 \times 5)^3$	1000

THERE IS NO SUCH LAW FOR A SUM

$(a + b)^2 \neq a^2 + b^2$. Try $a = b = 1$: $4 \neq 2$. The correct expansion is the subject of Quarter III.

Putting them together

EXPRESSION	WORKING	ANSWER
$(2x^2)^3$	$2^3(x^2)^3$	$8x^6$
$x^5 \cdot x^3 \div x^6$	x^{8-6}	x^2
$(a^3b)^2 \cdot a$	$a^6b^2 \cdot a$	a^7b^2
$\frac{(3a^2)^3}{9a^4}$	$\frac{27a^6}{9a^4}$	$3a^2$

DEAL WITH EACH BASE SEPARATELY

Collect the numbers, then the *as*, then the *bs*. Trying to do all three at once is where the errors come from.

02

Worked examples

EXAMPLE 1 Simplify $2^3 \cdot 2^4$, $3^7 \div 3^4$ and $(2^3)^2$.

1

Same base, multiply: add the exponents.

2

$2^7 = 128$

3

Same base, divide: subtract.

$3^3 = 27$.

4

Power of a power: multiply. $2^6 = 64$.

ANSWER

128, 27, 64

EXAMPLE 2 Simplify $(2x^2)^3$.

① Raise each factor: $2^3 \cdot (x^2)^3$.

② $2^3 = 8$

③ $(x^2)^3 = x^6$

Multiply the exponents.

④ $= 8x^6$

ANSWER

$8x^6$

EXAMPLE 3 Simplify $\frac{(3a^2)^3}{9a^4}$.

① Numerator: $27a^6$.

② Numbers: $\frac{27}{9} = 3$.

③ Letters: $a^6 \div a^4 = a^2$.

④ $= 3a^2$

ANSWER

$3a^2$

03 Interactive model

Write $a^3 \cdot a^2$ out in full as aaaaa before stating the law; the class sees that the rule is only counting.

Combining powers

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**

index n **3**

power m **1**

check: value³ **64**

a¹ **64**

Raising the answer to the power 3 gives back a¹ — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Law of indices	Daraja qonuni	Свойство степени
Same base	Bir xil asos	Одинаковое основание
To multiply powers	Darajalarni ko‘paytirish	Умножение степеней
To divide powers	Darajalarni bo‘lish	Деление степеней
Power of a power	Daraja darajasi	Степень степени
Power of a product	Ko‘paytma darajasi	Степень произведения
Exponent zero	Nolinch daraja	Нулевая степень
Simplify	Soddalashtirish	Упростить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $a^m \cdot a^n$ equals:

a^{mn}

a^{m+n}

a^{m-n}

$2a^m$

 $(a^m)^n$ equals:

a^{m+n}

a^{mn}

a^{m-n}

a^m

 $3^7 \div 3^4$ equals:

3^3

3^{11}

3^{28}

1

$(2x)^3$ equals:

$2x^3$

$6x^3$

$8x^3$

$8x$

 $(a + b)^2$ equals:

$a^2 + b^2$

$a^2 + 2ab + b^2$

$2a + 2b$

ab

 $a^6 \div a^6$ equals:

0

1

a

a^{12}

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up**7 PROBLEMS**

1. $2^3 \cdot 2^4$

ANSWER 2^7

2. $x^5 \cdot x$

ANSWER x^6

3. $3^7 \div 3^4$

ANSWER 3^3

4. $(2^3)^2$

ANSWER 2^6

5. $(x^4)^3$

ANSWER x^{12}

6. $(2x)^3$

ANSWER $8x^3$

7. $a^6 \div a^6$

ANSWER 1 **Medium · the standard question**

7 PROBLEMS

1. $(2x^2)^3$

ANSWER $8x^6$

2. $(-3a)^2$

ANSWER $9a^2$

3. $(\frac{x}{2})^4$

ANSWER $\frac{x^4}{16}$

4. $x^5 \cdot x^3 \div x^6$

ANSWER x^2

5. $(a^3b)^2 \cdot a$

ANSWER a^7b^2

6. $\frac{(3a^2)^3}{9a^4}$

ANSWER $3a^2$

7. $2^3 \cdot 3^2$

ANSWER 72 — bases differ

Hard · stretch and reason

7 PROBLEMS

1.
$$\frac{(2a^2b)^3}{4a^3b}$$

ANSWER $2a^3b^2$

2. $(x^2y^3)^2 \cdot (xy)^3$

ANSWER x^7y^9

3.
$$\frac{5^{n+1}}{5^n}$$

ANSWER 5

4. $(-2a^2)^3 \div (4a^3)$

ANSWER $-2a^3$

5. Is $(a + b)^2 = a^2 + b^2$?

ANSWER No — try $a = b = 1$

6. $2^{10} \div 2^7$

ANSWER 8

7. $(3x)^2 \cdot (2x)^3$

ANSWER $72x^5$

07 Homework

Homework — 5 tasks

Deal with the numbers first, then each letter separately.

1. Simplify $5^2 \cdot 5^3$, $7^6 \div 7^4$ and $(4^2)^3$.

2. Simplify $(3x^3)^2$ and $(-2a)^4$.

3. Simplify $y^7 \cdot y^2 \div y^5$.

4. Simplify $\frac{(2a^3)^4}{8a^5}$.

5. Show with numbers that $(a + b)^2 \neq a^2 + b^2$.

The monomial and its standard form

A number times some letters — the simplest object in algebra.

LESSON 65

1 × 40 MIN

MATEMATIKA 7, §29

S8 2.1

LESSON CLOCK

10'

12'

12'

6'

What a monomial is	10 min
Standard form	12 min
Degree and similarity	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define a monomial and recognise one.
- Write a monomial in standard form.
- Find the coefficient and the degree of a monomial.
- Recognise similar monomials.

TEXTBOOK

National	pp. 178–181
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

What a monomial is

A **monomial** is a product of numbers and letters, with natural-number exponents. No addition, no subtraction and no division by a letter.

EXPRESSION	A MONOMIAL?	WHY
$7a^2b$	yes	a product
$-x$	yes	$-1 \times x$
5	yes	a number alone
$a + b$	no	a sum
$\frac{3}{x}$	no	division by a letter
$\frac{x}{3}$	yes	$\frac{1}{3}x$

DIVIDING BY A NUMBER IS ALLOWED; DIVIDING BY A LETTER IS NOT

$\frac{x}{3}$ is the monomial $\frac{1}{3}x$; $\frac{3}{x}$ is not a monomial at all. The difference matters for everything that follows.

Standard form

A monomial is in **standard form** when it is written as one number followed by each letter once, in alphabetical order, with its exponent.

AS WRITTEN	STANDARD FORM
$3a \cdot 4b$	$12ab$
$x \cdot 5 \cdot x \cdot y$	$5x^2y$
$2b \cdot 3a \cdot a$	$6a^2b$
$-a \cdot a \cdot a \cdot 2$	$-2a^3$

MULTIPLY THE NUMBERS, COLLECT EACH LETTER

Two steps, always in that order. The alphabetical order is only a convention, but it makes two monomials easy to compare.

Degree and similarity

The **degree** of a monomial is the sum of the exponents of its letters. A number alone has degree 0.

MONOMIAL	COEFFICIENT	DEGREE
$7a^2b$	7	3
$-5x^3y^2$	-5	5
4	4	0
x	1	1

Two monomials are **similar** if their letter parts are identical — same letters, same exponents. Only similar monomials can be added.

PAIR	SIMILAR?
$3a^2b$ and $-7a^2b$	yes
$3a^2b$ and $3ab^2$	no
$5x$ and $5x^2$	no

A^2B AND AB^2 ARE NOT SIMILAR

Both have degree 3, but the letters carry different exponents. Similarity is about the letter part, not about the degree.

02

Worked examples

EXAMPLE 1 Write $2b \cdot 3a \cdot a$ in standard form, and give its coefficient and degree.

1

Numbers: $2 \times 3 = 6$.

2

Letters: $a \cdot a = a^2$, and one b .

3

$6a^2b$
Alphabetical order.

4

Coefficient 6, degree $2 + 1 = 3$.

ANSWER

$6a^2b$; 6 and 3

EXAMPLE 2 Which of $7a^2b$, $a + b$, $\frac{3}{x}$, $\frac{x}{3}$, 5 are monomials?

- ① $7a^2b$ — a product ✓
- ② $a + b$ — a sum ✗
- ③ $\frac{3}{x}$ — division by a letter ✗
- ④ $\frac{x}{3}$ and 5 ✓

ANSWER

$7a^2b$, $\frac{x}{3}$, 5

EXAMPLE 3 Are $3a^2b$ and $3ab^2$ similar? And $3a^2b$ and $-7a^2b$?

- ① Compare the letter parts.
- ② a^2b and ab^2 differ.
Not similar.
- ③ a^2b and a^2b agree.
- ④ Similar — only the coefficients differ.

ANSWER

No; yes

03 Interactive model

Write the same monomial six different ways on the board and ask which is “tidiest”; the class defines standard form before it is named.

This model could not start: den is not a function

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Monomial	Bir had	Одночлен
Standard form	Standart ko‘rinish	Стандартный вид
Coefficient	Koeffitsiyent	Коэффициент
Degree of a monomial	Bir hadning darajasi	Степень одночлена
Similar monomials	O‘xshash bir hadlar	Подобные одночлены
Letter part	Harfiy qism	Буквенная часть
Alphabetical order	Alifbo tartibi	Алфавитный порядок
Zero monomial	Nol bir had	Нулевой одночлен

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Which is a monomial?

$$a + b$$

$$7a^2b$$

$$\frac{3}{x}$$

$$a - 1$$

Is x 3 a monomial?

yes

no

sometimes

only if $x > 0$

Standard form of $2b \cdot 3a \cdot a$:

$$6ab^2$$

$$6a^2b$$

$$5a^2b$$

$$6a^3$$

The degree of $-5x^3y^2$:

The degree of 4:

Are $3a^2b$ and $3ab^2$ similar?

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Is $7a^2b$ a monomial?

ANSWER Yes

2. Is $a + b$ a monomial?

ANSWER No

3. Is $\frac{3}{x}$ a monomial?

ANSWER No

4. Is 5 a monomial?

ANSWER Yes

5. Coefficient of $7a^2b$

ANSWER 7

6. Degree of $7a^2b$

ANSWER 3

7. Degree of 4

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of $3a \cdot 4b$

ANSWER $12ab$

2. Standard form of $x \cdot 5 \cdot x \cdot y$

ANSWER $5x^2y$

3. Standard form of $2b \cdot 3a \cdot a$

ANSWER $6a^2b$

4. Coefficient and degree of $-5x^3y^2$

ANSWER -5 and 5

5. Are $3a^2b$ and $-7a^2b$ similar?

ANSWER Yes

6. Are $5x$ and $5x^2$ similar?

ANSWER No

7. Standard form of $-a \cdot a \cdot a \cdot 2$

ANSWER $-2a^3$

Hard · stretch and reason

7 PROBLEMS

1. Standard form of $(2a)^3 \cdot 3b$

ANSWER $24a^3b$

2. Degree of $(3x^2y)^2$

ANSWER 6

3. Standard form of $\frac{1}{2}x \cdot 4x^2y$

ANSWER $2x^3y$

4. Two similar monomials of degree 4

ANSWER $3a^2b^2$ and $-a^2b^2$

5. Is x^{-1} a monomial?

ANSWER No — the exponent is not natural

6. Degree of $7a^2b \cdot 3ab^3$

ANSWER 7

7. Coefficient of $-\frac{2x^3}{5}$

ANSWER $-\frac{2}{5}$

07 Homework

Homework — 5 tasks

Write every monomial in standard form before giving its coefficient or degree.

- Which of $4xy$, $x + 3$, $\frac{5}{y}$, $-2a^3$, 7 are monomials?
- Write $5a \cdot 2b \cdot a$ in standard form.
- Give the coefficient and degree of $-3x^2y^4$.

4. Are $4ab^2$ and $-ab^2$ similar? And $4ab^2$ and $4a^2b$?
5. Write $(3x)^2 \cdot 2y$ in standard form and give its degree.

Multiplication and division of monomials

Numbers with numbers, letters with letters — and the laws of indices do the rest.

LESSON 66–67

2 × 40 MIN

MATEMATIKA 7, §30

S8 2.1

LESSON CLOCK

13'

23'

22'

18'

4'

Multiplying	13 min
Raising to a power	23 min
Dividing	22 min
Areas and volumes	18 min
Homework	4 min

LEARNING OBJECTIVES

- Multiply two or more monomials and give the answer in standard form.
- Divide one monomial by another when the division is exact.
- Raise a monomial to a power.
- Use the operations in area and volume problems.

TEXTBOOK

National	pp. 182–188
Cambridge	Stage 8, pp. 12–18
Workbook	Exercise 2.1

01

Explanation

Multiplying

Multiply the coefficients; then, for each letter, add the exponents.

PRODUCT	NUMBERS	LETTERS	ANSWER
$3a \cdot 4b$	12	ab	$12ab$
$2x^2 \cdot 5x^3$	10	x^5	$10x^5$
$(-3a^2b)(4ab^3)$	-12	a^3b^4	$-12a^3b^4$
$(\frac{1}{2}xy)(6x^2)$	3	x^3y	$3x^3y$

THE PRODUCT OF TWO MONOMIALS IS ALWAYS A MONOMIAL

Its degree is the sum of the two degrees. That is worth checking as you go: $(-3a^2b)$ has degree 3, $(4ab^3)$ degree 4, and the answer degree 7 ✓

Raising to a power

$$(ka^mb^n)^p = k^p a^{mp} b^{np}$$

Every factor is raised — the coefficient included.

EXPRESSION	ANSWER
$(2x)^3$	$8x^3$
$(3a^2b)^2$	$9a^4b^2$
$(-2x^3)^3$	$-8x^9$
$(-2x^3)^2$	$4x^6$

THE COEFFICIENT IS RAISED TOO

$(3a^2b)^2 = 9a^4b^2$, not $3a^4b^2$. Forgetting to square the 3 is the standard slip.

Dividing

Divide the coefficients; then, for each letter, subtract the exponents.

QUOTIENT	NUMBERS	LETTERS	ANSWER
$12a^5 \div 4a^2$	3	a^3	$3a^3$
$-15x^3y^2 \div 5xy$	-3	x^2y	$-3x^2y$
$8a^4b^3 \div (-2a^4b)$	-4	b^2	$-4b^2$

THE QUOTIENT IS A MONOMIAL ONLY IF EVERY EXPONENT STAYS NATURAL

$a^2 \div a^5$ is not a monomial. In Grade 7 the divisions set are always exact, but the condition is worth knowing.

Areas and volumes

FIGURE	DIMENSIONS	AREA OR VOLUME
rectangle	$3a$	$12ab$
square	$2x$	$4x^2$
cuboid	$a, 2a, 3a$	$6a^3$
cube	$2x$	$8x^3$

The first row assumes a rectangle $3a$ by $4b$. Notice that the units follow the degree: a length is degree 1, an area degree 2, a volume degree 3.

THE DEGREE IS A DIMENSION CHECK

An answer of degree 2 for a volume is wrong before any arithmetic is examined. This is the algebraic version of checking the units.

02

Worked examples

EXAMPLE 1 Simplify $(-3a^2b)(4ab^3)$.

1

Numbers: $-3 \times 4 = -12$.

2

a : $a^2 \cdot a = a^3$.

3

b : $b \cdot b^3 = b^4$.

4

$= -12a^3b^4$
Degree 7 ✓

ANSWER

$-12a^3b^4$

EXAMPLE 2 Simplify $(3a^2b)^2$ and $(-2x^3)^3$.

① Square every factor: $3^2 = 9$, $(a^2)^2 = a^4$, b^2 .

② $= 9a^4b^2$

③ Cube every factor: $(-2)^3 = -8$, $(x^3)^3 = x^9$.

④ $= -8x^9$

ANSWER $9a^4b^2$ and $-8x^9$ **EXAMPLE 3** Simplify $-15x^3y^2 \div 5xy$.

① Numbers: $-15 \div 5 = -3$.

② x : $x^3 \div x = x^2$.

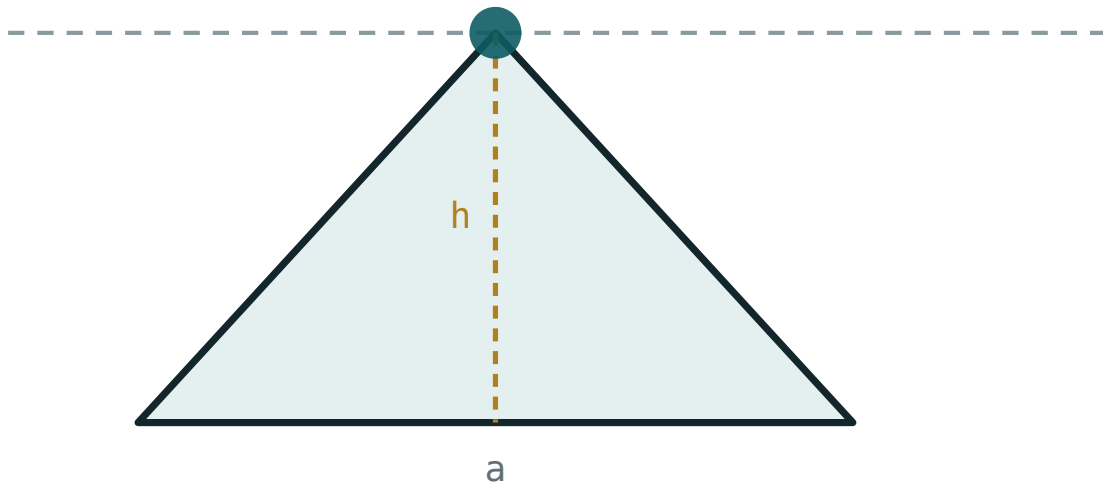
③ y : $y^2 \div y = y$.

④ $= -3x^2y$

ANSWER $-3x^2y$ **03 Interactive model**

Draw a rectangle $3a$ by $4b$ on the board and ask for its area before any rule is stated; the class multiplies numbers and letters separately without being told to.

A rectangle with algebraic sides

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply	Ko'paytirish	Умножить
To divide	Bo'lish	Разделить
Standard form	Standart ko'rinish	Стандартный вид
Exactly divisible	Butun bo'linadi	Делится нацело
Power of a monomial	Bir had darajasi	Степень одночлена
Area	Yuza	Площадь
Volume	Hajm	Объём
Cancel	Qisqartirish	Сократить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself** $2x^2 \cdot 5x^3$ equals:

$10x^5$

$10x^6$

$7x^5$

$10x$

 $(3a^2b)^2$ equals:

$3a^4b^2$

$9a^4b^2$

$9a^4b$

$6a^4b^2$

 $(-2x^3)^3$ equals:

$-8x^9$

$8x^9$

$-6x^9$

$-8x^6$

$12a^5 \div 4a^2$ equals:

The degree of a product of monomials is:

A cube of side $2x$ has volume:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3a \cdot 4b$

ANSWER $12ab$

2. $2x^2 \cdot 5x^3$

ANSWER $10x^5$

3. $(2x)^3$

ANSWER $8x^3$

4. $(3a^2b)^2$

ANSWER $9a^4b^2$

5. $12a^5 \div 4a^2$

ANSWER $3a^3$

6. $(-2x^3)^2$

ANSWER $4x^6$

7. $(-2x^3)^3$

ANSWER $-8x^9$ **Medium · the standard question**

7 PROBLEMS

1. $(-3a^2b)(4ab^3)$

ANSWER $-12a^3b^4$

2. $(\frac{1}{2}xy)(6x^2)$

ANSWER $3x^3y$

3. $-15x^3y^2 \div 5xy$

ANSWER $-3x^2y$

4. $8a^4b^3 \div (-2a^4b)$

ANSWER $-4b^2$

5. Area of a rectangle $3a$ by $4b$

ANSWER $12ab$

6. Volume of a cuboid a , $2a$, $3a$

ANSWER $6a^3$

7. Degree of $(-3a^2b)(4ab^3)$

ANSWER 7

Hard · stretch and reason

7 PROBLEMS

1. $(2a^2b)^3 \cdot 3ab$

ANSWER $24a^7b^4$

2. $\frac{(3xy^2)^3}{9x^2y^4}$

ANSWER $3xy^2$

3. $(-a^2b)^4 \div (a^3b^2)$

ANSWER a^5b^2

4. A cube of volume $27x^6$: its edge

ANSWER $3x^2$

5. A square of area $49a^4$: its side

ANSWER $7a^2$

6. $(2x)^2 \cdot (3x)^3$

ANSWER $108x^5$

7. Why is $a^2 \div a^5$ not a monomial?

ANSWER The exponent would be negative

07 Homework**Homework — 5 tasks**

Check the degree of every answer against the degrees you started with.

1. Simplify $(-2a^3b)(5ab^2)$.
2. Simplify $(4x^2y)^2$ and $(-3a)^3$.
3. Simplify $20a^6b^3 \div 5a^2b$.
4. Find the volume of a cuboid with edges $2a$, $3a$ and $4b$.

5. A cube has volume $64x^3$. Find its edge.

Polynomials

A sum of monomials — and the object that the rest of algebra is about.

LESSON 68

1 × 40 MIN

MATEMATIKA 7, §31

S8 2.2

LESSON CLOCK

10'

12'

12'

6'

The definition	10 min
Standard form and degree	12 min
Evaluating	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define a polynomial and name its terms.
- Write a polynomial in standard form and give its degree.
- Name a binomial and a trinomial.
- Evaluate a polynomial for given values.

TEXTBOOK

National	pp. 189–192
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

The definition

A **polynomial** is a sum of monomials. Each monomial is a **term**.

POLYNOMIAL	TERMS	NAME
$3x$	1	monomial
$3x + 5$	2	binomial
$x^2 - 4x + 3$	3	trinomial
$a^3 + a^2 - a + 7$	4	polynomial

Subtraction counts as addition of a negative, so $x^2 - 4x$ is the sum of x^2 and $-4x$.

EVERY MONOMIAL IS A POLYNOMIAL

With one term. The names simply record how many terms there are, and only the first three have special names.

Standard form and degree

A polynomial is in **standard form** when every term is a monomial in standard form, no two terms are similar, and the terms are written in decreasing degree.

AS WRITTEN	STANDARD FORM	DEGREE
$5 + x^2 - 3x$	$x^2 - 3x + 5$	2
$2x + 3x$	$5x$	1
$a^3 + 2a^3 - 4$	$3a^3 - 4$	3
$x^2y + xy^2 + 1$	already standard	3

The **degree of a polynomial** is the greatest of the degrees of its terms. The term of that degree is the **leading term**; a term with no letters is the **constant term**.

COLLECT SIMILAR TERMS BEFORE READING THE DEGREE

In $a^3 + 2a^3 - 4$ the first two terms combine. Reading the degree from an uncollected expression can give the wrong answer when terms cancel.

Evaluating

Substitute and compute, with every negative value in brackets.

POLYNOMIAL	$X = 2$	$X = -1$
$x^2 - 4x + 3$	$4 - 8 + 3 = -1$	$1 + 4 + 3 = 8$
$2x^3 - x$	$16 - 2 = 14$	$-2 + 1 = -1$
$5 - x^2$	$5 - 4 = 1$	$5 - 1 = 4$

A VALUE THAT MAKES THE POLYNOMIAL ZERO IS A ROOT

$x^2 - 4x + 3$ is zero at $x = 1$ and at $x = 3$. Finding roots is what the whole of the equation work later this year is about.

02 Worked examples**EXAMPLE 1** Write $5 + x^2 - 3x$ in standard form and give its degree and constant term.

- ① Order by decreasing degree.

- ② $x^2 - 3x + 5$

- ③ Degree 2 — the greatest term degree.

- ④ Constant term 5.

ANSWER $x^2 - 3x + 5$; degree 2, constant 5**EXAMPLE 2** Simplify $a^3 + 2a^3 - 4$ and give its degree.

- ① Collect the similar terms.

- ② $3a^3 - 4$

- ③ Two terms — a binomial.

- ④ Degree 3.

ANSWER $3a^3 - 4$, degree 3

EXAMPLE 3 Evaluate $x^2 - 4x + 3$ at $x = 2$ and at $x = -1$.

① $4 - 8 + 3$

② $= -1$

③ $(-1)^2 - 4(-1) + 3 = 1 + 4 + 3$

Brackets round the negative.

④ $= 8$

ANSWER

-1 and 8

03 Interactive model

Ask for the value of $x^2 - 4x + 3$ at $x = 1$ and $x = 3$; both give zero, and the idea of a root arrives before any equation is written.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Polynomial	Ko'phad	Многочлен
Binomial	Ikki had	Двучлен
Trinomial	Uch had	Трёхчлен
Term of a polynomial	Ko'phad hadi	Член многочлена
Degree of a polynomial	Ko'phad darajasi	Степень многочлена
Standard form	Standart ko'rinish	Стандартный вид
Leading term	Bosh had	Старший член
Constant term	Ozod had	Свободный член

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A polynomial is:

a product of monomials

a sum of monomials

a fraction

an equation

$3x + 5$ is a:

monomial

binomial

trinomial

constant

The degree of $x^2 - 3x + 5$:

1

2

3

5

The constant term of $x^2 - 3x + 5$:

$a^3 + 2a^3 - 4$ simplifies to:

$x^2 - 4x + 3$ at $x = -1$:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. How many terms in $3x + 5$?

ANSWER 2

2. Its name

ANSWER Binomial

3. How many terms in $x^2 - 4x + 3$?

ANSWER 3

4. Its name

ANSWER Trinomial

5. Degree of $x^2 - 3x + 5$

ANSWER 2

6. Constant term of $x^2 - 3x + 5$

ANSWER 5

7. Degree of 7

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. Standard form of $5 + x^2 - 3x$

ANSWER $x^2 - 3x + 5$

2. Simplify $2x + 3x$

ANSWER $5x$

3. Simplify $a^3 + 2a^3 - 4$

ANSWER $3a^3 - 4$

4. Degree of $x^2y + xy^2 + 1$

ANSWER 3

5. $x^2 - 4x + 3$ at $x = 2$

ANSWER -1

6. $x^2 - 4x + 3$ at $x = -1$

ANSWER 8

7. $2x^3 - x$ at $x = 2$

ANSWER 14

Hard · stretch and reason

7 PROBLEMS

1. For which x is $x^2 - 4x + 3 = 0$?

ANSWER $x = 1$ and $x = 3$

2. Simplify $3x^2 - 5x + 2 - x^2 + 5x - 7$

ANSWER $2x^2 - 5$

3. Its degree

ANSWER 2

4. Degree of $(x + 1)(x - 1)$

ANSWER 2

5. $5 - x^2$ at $x = -3$

ANSWER -4

6. A polynomial of degree 0

ANSWER A non-zero constant

7. Simplify $a^2 + a^2 + a^2$

ANSWER $3a^2$

07 Homework

Homework — 5 tasks

Collect similar terms before naming the degree.

1. Write $7 - 2x + x^3$ in standard form and give its degree.
2. Simplify $4y^2 + 3y^2 - 5$ and name the result.
3. Give the constant term of $2x^2 - x + 9$.
4. Evaluate $x^2 + 2x - 5$ at $x = 3$ and at $x = -2$.
5. Find the value of x for which $3x - 6 = 0$.

Like terms and their collection

Only terms with the same letters can be added — and that one rule tidies every expression.

LESSON 69–70

2 × 40 MIN

MATEMATIKA 7, §32

S8 2.2

LESSON CLOCK

13'

23'

22'

18'

4'

Which terms are alike	13 min
Collecting	23 min
After brackets	22 min
In problems	18 min
Homework	4 min

LEARNING OBJECTIVES

- Recognise like terms.
- Collect like terms, including with negative coefficients.
- Simplify an expression after removing brackets.
- Use collection to shorten a perimeter or a cost calculation.

TEXTBOOK

National	pp. 193–198
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

Which terms are alike

Two terms are **like** if their letter parts are identical — the same letters with the same exponents. The coefficients may be anything.

PAIR	LIKE?	WHY
$3x, -7x$	yes	same letter part x
$4a^2b, a^2b$	yes	same letter part
$4a^2b, 4ab^2$	no	different exponents
$5x, 5y$	no	different letters
$7, -2$	yes	both constants

THINK OF THE LETTER PART AS A UNIT

Three apples plus five apples is eight apples; three apples plus five pears is neither. That is the whole rule, and it explains why unlike terms simply stay as they are.

Collecting

$$ka + la = (k + l)a$$

Add the coefficients and keep the letter part.

EXPRESSION	COLLECTED
$3x + 5x$	$8x$
$7a - 10a$	$-3a$
$4x + 3y - x + 5y$	$3x + 8y$
$2a^2 + 3a - a^2 + a$	$a^2 + 4a$

TAKE THE SIGN WITH THE TERM

In $4x + 3y - x + 5y$ the third term is $-x$, so the x terms give $4 - 1 = 3$. Copying a term without its sign is the commonest error here.

After brackets

Remove the brackets first, then collect.

EXPRESSION	BRACKETS REMOVED	COLLECTED
$2(x + 3) + 3(x - 1)$	$2x + 6 + 3x - 3$	$5x + 3$
$5(a - 2) - 2(a - 5)$	$5a - 10 - 2a + 10$	$3a$
$x - (3x - 4)$	$x - 3x + 4$	$-2x + 4$

TWO PASSES, NEVER ONE

Removing brackets and collecting at the same time is where the signs go wrong. Write the expanded line in full before touching the collection.

In problems

SITUATION	EXPRESSION	SIMPLIFIED
perimeter of a rectangle a by b	$a + b + a + b$	$2a + 2b$
perimeter of a triangle x , x , $2x - 1$	$x + x + 2x - 1$	$4x - 1$
3 pens at p and 4 books at b , twice over	$2(3p + 4b)$	$6p + 8b$

SIMPLIFYING IS WHAT MAKES A FORMULA USABLE

$2a + 2b$ can be evaluated in two operations; $a + b + a + b$ needs three. Over a whole calculation the saving is large, and the shorter form is also easier to read.

02 Worked examples

EXAMPLE 1 Collect $4x + 3y - x + 5y$.

- 1

x terms: $4x - x = 3x$.

The sign belongs to the term.
- 2

y terms: $3y + 5y = 8y$.
- 3

They cannot be combined further.
- 4

$3x + 8y$

ANSWER

$3x + 8y$

EXAMPLE 2 Simplify $5(a - 2) - 2(a - 5)$.

- ① $5a - 10$ and $-2a + 10$.
Both signs change in the second.

② $5a - 10 - 2a + 10$

③ a terms: $3a$; constants: 0 .

④ $= 3a$

ANSWER

$$3a$$

EXAMPLE 3 Find and simplify the perimeter of a triangle with sides x , x and $2x - 1$.

① $P = x + x + (2x - 1)$

② $= x + x + 2x - 1$

③ Collect: $4x - 1$.

④ Check at $x = 3$: $3 + 3 + 5 = 11 = 4(3) - 1$ ✓

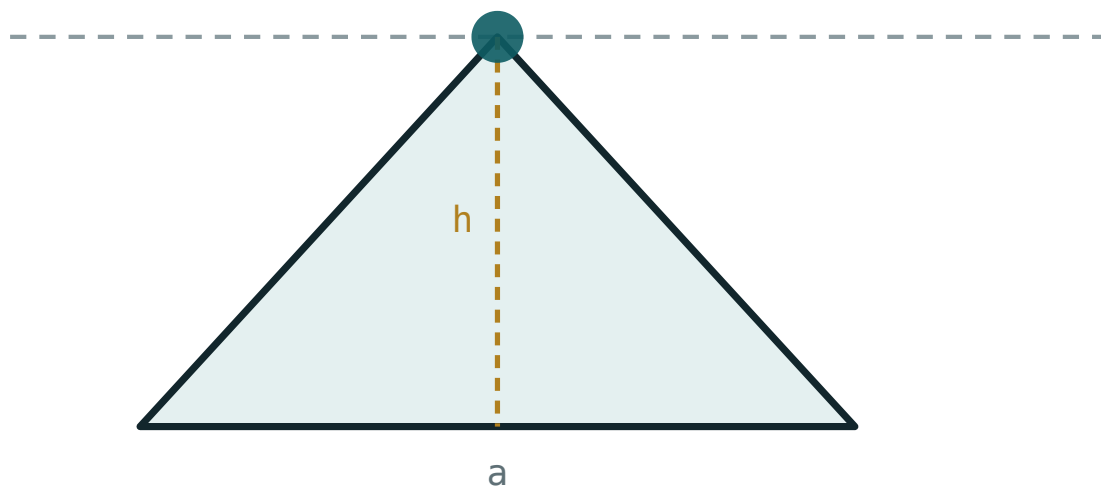
ANSWER

$$4x - 1$$

03 Interactive model

Ask for the perimeter of a rectangle twice — once unsimplified and once collected — and have the class evaluate both at $a = 7$, $b = 4$; the answers agree and one is far quicker.

Collecting like terms

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Like terms	O'xshash hadlar	Подобные члены
To collect	Ixchamlash	Приведение подобных
Unlike terms	O'xshash bo'lmagan hadlar	Неподобные члены
Coefficient	Koeffitsiyent	Коэффициент
Letter part	Harfiy qism	Буквенная часть
To simplify	Soddalashtirish	Упростить
Perimeter	Perimetr	Периметр
Cost	Narx	Стоимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Are $4a^2b$ and $4ab^2$ like terms?

$3x + 5x$ equals:

$7a - 10a$ equals:

$4x + 3y - x + 5y$ equals:

$3x + 8y$

$5x + 8y$

$3x + 2y$

$11xy$

$5(a - 2) - 2(a - 5)$ equals:

$3a - 20$

$3a$

$7a - 20$

$3a + 20$

Can $3x$ and $5y$ be collected?

yes

no

as $8xy$

as $15xy$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3x + 5x$

ANSWER $8x$

2. $7a - 10a$

ANSWER $-3a$

3. $2b + b$

ANSWER $3b$

4. $5x - 5x$

ANSWER 0

5. $4a^2 + 3a^2$

ANSWER $7a^2$

6. Can $3x$ and $5y$ be collected?

ANSWER No

7. Are 7 and -2 like terms?

ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. $4x + 3y - x + 5y$

ANSWER $3x + 8y$

2. $2a^2 + 3a - a^2 + a$

ANSWER $a^2 + 4a$

3. $2(x + 3) + 3(x - 1)$

ANSWER $5x + 3$

4. $5(a - 2) - 2(a - 5)$

ANSWER $3a$

5. $x - (3x - 4)$

ANSWER $4 - 2x$

6. Perimeter of a rectangle a by b

ANSWER $2a + 2b$

7. Perimeter of a triangle $x, x, 2x - 1$

ANSWER $4x - 1$

Hard · stretch and reason

7 PROBLEMS

1. $3(2a - b) - 2(3a - 2b)$

ANSWER b

2. $4x^2 - 3x + 1 - (x^2 - 3x + 5)$

ANSWER $3x^2 - 4$

3. $a - [2a - (3a - b)]$

ANSWER $2a - b$

4. $2(x + y) - 3(x - y) + x$

ANSWER $5y$

5. Perimeter of a square of side $3x - 2$

ANSWER $12x - 8$

6. A rectangle $x + 3$ by $x - 3$: its perimeter

ANSWER $4x$

7. Simplify $5ab - 2ba$

ANSWER $3ab$

07 Homework

Homework — 5 tasks

Write the expanded line in full before collecting anything.

1. Collect $6a + 2b - 4a + 7b$.

2. Simplify $3(x + 4) - 2(x - 1)$.

3. Simplify $y - (4y - 7)$.

4. Find and simplify the perimeter of a rectangle $2x + 1$ by $x - 2$.

5. Simplify $5m^2 + 2m - 3m^2 - 6m$.

Addition and subtraction of polynomials

Remove the brackets, collect the like terms — two steps and nothing more.

LESSON 71–72

2 × 40 MIN

MATEMATIKA 7, §33

S8 2.2

LESSON CLOCK

13'

23'

22'

18'

4'

Adding	13 min
Subtracting	23 min
Longer expressions	22 min
Checking	18 min
Homework	4 min

LEARNING OBJECTIVES

- Add two polynomials.
- Subtract one polynomial from another, changing every sign.
- Simplify an expression containing several polynomials.
- Check an answer by substituting a value.

TEXTBOOK

National	pp. 199–204
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01

Explanation

Adding

A plus before a bracket leaves every sign as it is.

$$(3x^2 - 2x + 5) + (x^2 + 4x - 1) = 4x^2 + 2x + 4$$

STEP	LINE
remove brackets	$3x^2 - 2x + 5 + x^2 + 4x - 1$
x^2	$3x^2 + x^2 = 4x^2$
x	$-2x + 4x = 2x$
constants	$5 - 1 = 4$

WORK ONE DEGREE AT A TIME

All the x^2 terms, then all the x terms, then the constants. Doing them in a fixed order means nothing is missed.

Subtracting

A minus before a bracket changes **every** sign inside it.

$$(3x^2 - 2x + 5) - (x^2 + 4x - 1) = 2x^2 - 6x + 6$$

STEP	LINE
remove brackets	$3x^2 - 2x + 5 - x^2 - 4x + 1$
x^2	$3x^2 - x^2 = 2x^2$
x	$-2x - 4x = -6x$
constants	$5 + 1 = 6$

ALL THREE SIGNS CHANGE, NOT JUST THE FIRST

$-(x^2 + 4x - 1)$ is $-x^2 - 4x + 1$. Writing that line out in full, before collecting, is the single most useful habit in this topic.

Longer expressions

EXPRESSION	ANSWER
$(a + b) + (a - b)$	$2a$
$(a + b) - (a - b)$	$2b$
$(2x - 3) + (x + 1) - (x - 4)$	$2x + 2$
$2(x^2 - x) - 3(x^2 + x)$	$-x^2 - 5x$

The last row combines this lesson with the distributive law: expand first, then collect.

THE DEGREE NEVER RISES

Adding or subtracting polynomials of degree 2 gives a polynomial of degree at most 2 — and sometimes less, if the leading terms cancel.

Checking

Substitute one convenient value into the original and into the answer. If they differ, the simplification is wrong.

CHECK	ORIGINAL AT $x = 1$	ANSWER AT $x = 1$
$(3x^2 - 2x + 5) + (x^2 + 4x - 1)$	$6 + 4 = 10$	$4 + 2 + 4 = 10 \checkmark$
$(3x^2 - 2x + 5) - (x^2 + 4x - 1)$	$6 - 4 = 2$	$2 - 6 + 6 = 2 \checkmark$

USE $x = 1$, OR $x = 2$ IF TERMS CANCEL

At $x = 1$ every power is 1, so the check is a sum of the coefficients — the quickest arithmetic there is. If it happens to give 0 on both sides by accident, try $x = 2$ as well.

02

Worked examples

EXAMPLE 1 Compute $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$.

1

A plus: signs unchanged.

2

$3x^2 - 2x + 5 + x^2 + 4x - 1$

3

Collect by degree.

4

$= 4x^2 + 2x + 4$
Check at $x = 1$: $10 = 10 \checkmark$

ANSWER

$4x^2 + 2x + 4$

EXAMPLE 2 Compute $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$.

1 A minus: every sign in the second bracket changes.

2 $3x^2 - 2x + 5 - x^2 - 4x + 1$

3 Collect by degree.

4 $= 2x^2 - 6x + 6$

Check at $x = 1$: $2 = 2$ ✓

ANSWER

$$2x^2 - 6x + 6$$

EXAMPLE 3 Simplify $2(x^2 - x) - 3(x^2 + x)$.

1 Expand: $2x^2 - 2x$ and $-3x^2 - 3x$.

2 $2x^2 - 2x - 3x^2 - 3x$

3 x^2 : $-x^2$; x : $-5x$.

4 $= -x^2 - 5x$

ANSWER

$$-x^2 - 5x$$

03 Interactive model

Do one subtraction wrongly on purpose — changing only the first sign — and check it at $x = 1$; the check fails, and the class sees why the habit matters.

This model could not start: den is not a function

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Sum of polynomials	Ko‘phadlar yig‘indisi	Сумма многочленов
Difference	Ayirma	Разность
To remove brackets	Qavslarni ochish	Раскрыть скобки
Like terms	O‘xshash hadlar	Подобные члены
Opposite polynomial	Qarama-qarshi ko‘phad	Противоположный многочлен
Check by substitution	O‘rniga qo‘yib tekshirish	Проверка подстановкой
Standard form	Standart ko‘rinish	Стандартный вид
Degree	Daraja	Степень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A plus before a bracket:

changes all signs

changes none

changes the first

removes the terms

A minus before a bracket:

changes all signs

changes none

changes the first

doubles the terms

$(a + b) - (a - b)$ equals:

$2a$

$2b$

0

$2a + 2b$

$(3x^2 - 2x + 5) - (x^2 + 4x - 1)$ equals:

The best value for a quick check:

Adding two polynomials of degree 2 gives degree:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(a + b) + (a - b)$

ANSWER $2a$

2. $(a + b) - (a - b)$

ANSWER $2b$

3. $(x + 2) + (x + 3)$

ANSWER $2x + 5$

4. $(x + 2) - (x + 3)$

ANSWER -1

5. $(2x - 1) + (3x + 4)$

ANSWER $5x + 3$

6. $(2x - 1) - (3x + 4)$

ANSWER $-x - 5$

7. A minus before a bracket

ANSWER Changes every sign

Medium · the standard question

7 PROBLEMS

1. $(3x^2 - 2x + 5) + (x^2 + 4x - 1)$

ANSWER $4x^2 + 2x + 4$

2. $(3x^2 - 2x + 5) - (x^2 + 4x - 1)$

ANSWER $2x^2 - 6x + 6$

3. $(2x - 3) + (x + 1) - (x - 4)$

ANSWER $2x + 2$

4. $2(x^2 - x) - 3(x^2 + x)$

ANSWER $-x^2 - 5x$

5. $(a^2 + ab) - (ab - b^2)$

ANSWER $a^2 + b^2$

6. $(5m - 2n) - (3m - 7n)$

ANSWER $2m + 5n$

7. Check $4x^2 + 2x + 4$ at $x = 1$

ANSWER 10

Hard · stretch and reason

7 PROBLEMS

1. $(x^3 - 2x^2 + x) - (x^3 + x^2 - 3x)$

ANSWER $-3x^2 + 4x$

2. $3(2a - b) - 2(a - 3b) + (a - b)$

ANSWER $5a + 2b$

3. Subtract $x^2 - 3x + 2$ from $2x^2 + x - 5$

ANSWER $x^2 + 4x - 7$

4. What must be added to $3x - 4$ to give $5x + 1$?

ANSWER $2x + 5$

5. $(a + b + c) - (a - b + c)$

ANSWER $2b$

6. The degree of $(x^2 + x) - (x^2 - x)$

ANSWER 1

7. Simplify $(2x^2 - 3) + (3 - 2x^2)$

ANSWER 0

07 Homework

Homework — 5 tasks

Write the bracket-free line in full, then collect, then check at $x = 1$.

1. Compute $(4x^2 + x - 3) + (2x^2 - 5x + 7)$.
2. Compute $(4x^2 + x - 3) - (2x^2 - 5x + 7)$.
3. Simplify $3(a - 2b) - 2(2a - b)$.
4. Subtract $2y^2 - y + 4$ from $5y^2 + 3y - 1$.
5. What must be added to $7m - 2$ to give $4m + 5$?

Multiplication of polynomials

Every term of the first times every term of the second — and a rectangle explains why.

LESSON 73–74

2 × 40 MIN

MATEMATIKA 7, §34

S8 2.3

LESSON CLOCK

13'

23'

22'

18'

4'

By a monomial	13 min
Two binomials	23 min
The area picture	22 min
Longer products	18 min
Homework	4 min

LEARNING OBJECTIVES

- Multiply a polynomial by a monomial.
- Multiply two binomials and collect the result.
- Multiply a binomial by a trinomial.
- Use an area diagram to justify the expansion.

TEXTBOOK

National	pp. 205–211
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

By a monomial

$$k(a + b + c) = ka + kb + kc$$

Multiply every term — including the last, and including the signs.

PRODUCT	EXPANDED
$3x(x - 4)$	$3x^2 - 12x$
$-2a(a^2 - 3a + 1)$	$-2a^3 + 6a^2 - 2a$
$xy(x + y)$	$x^2y + xy^2$

THE LAST TERM IS THE ONE MOST OFTEN FORGOTTEN

In $-2a(a^2 - 3a + 1)$ the third term gives $-2a$. Counting the terms before and after expanding catches the omission.

Two binomials

$$(a + b)(c + d) = ac + ad + bc + bd$$

Four products, one for each pair. Then collect any like terms.

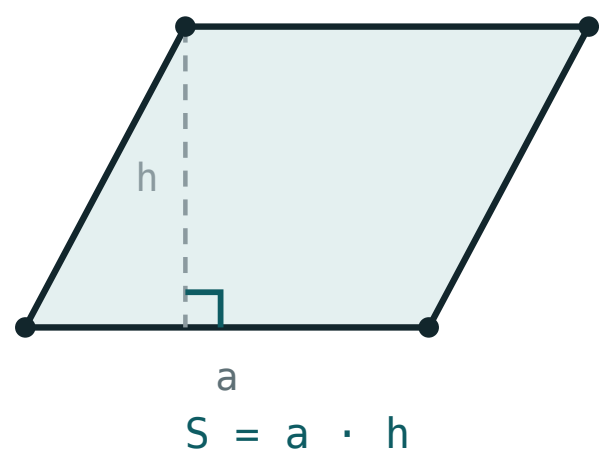
PRODUCT	FOUR TERMS	COLLECTED
$(x + 2)(x + 3)$	$x^2 + 3x + 2x + 6$	$x^2 + 5x + 6$
$(x - 2)(x + 5)$	$x^2 + 5x - 2x - 10$	$x^2 + 3x - 10$
$(2a - 1)(a + 4)$	$2a^2 + 8a - a - 4$	$2a^2 + 7a - 4$
$(x - 3)(x + 3)$	$x^2 + 3x - 3x - 9$	$x^2 - 9$

THE LAST ROW IS WORTH REMEMBERING

The middle terms cancel and only $x^2 - 9$ survives. That pattern — the difference of two squares — is the subject of a whole lesson in Quarter III.

The area picture

A rectangle of sides $a + b$ and $c + d$ splits into four smaller rectangles.



A rectangle cut into four — the four products of the expansion.

	C	D
a	ac	ad
b	bc	bd

The whole area is the sum of the four parts — which is exactly the expansion. The grid also makes it impossible to miss a product.

USE THE GRID WHEN THERE ARE MORE THAN FOUR PRODUCTS

For a binomial times a trinomial there are six, and a 2×3 grid keeps them all in view.

Longer products

Example. $(x + 2)(x^2 - 3x + 1)$. Six products:

	X^2	$-3X$	$+1$
x	x^3	$-3x^2$	x
$+2$	$2x^2$	$-6x$	2

$$x^3 - 3x^2 + 2x^2 + x - 6x + 2 = x^3 - x^2 - 5x + 2$$

THE DEGREE OF A PRODUCT IS THE SUM OF THE DEGREES

Here $1 + 2 = 3$, and the answer is cubic. Checking the degree first tells you what to expect before any arithmetic.

02 Worked examples

EXAMPLE 1 Expand $-2a(a^2 - 3a + 1)$.

$$① \quad -2a \times a^2 = -2a^3$$

$$② \quad -2a \times (-3a) = 6a^2$$

Two minuses.

$$③ \quad -2a \times 1 = -2a$$

$$④ \quad = -2a^3 + 6a^2 - 2a$$

Three terms in, three out.

ANSWER

$$-2a^3 + 6a^2 - 2a$$

EXAMPLE 2 Expand $(x - 2)(x + 5)$.

$$① \quad x \cdot x = x^2, x \cdot 5 = 5x.$$

$$② \quad -2 \cdot x = -2x, -2 \cdot 5 = -10.$$

$$③ \quad x^2 + 5x - 2x - 10$$

$$④ \quad = x^2 + 3x - 10$$

ANSWER

$$x^2 + 3x - 10$$

EXAMPLE 3 Expand $(x + 2)(x^2 - 3x + 1)$.

① x row: $x^3 - 3x^2 + x$.

② $+2$ row: $2x^2 - 6x + 2$.

③ Add and collect.

④ $= x^3 - x^2 - 5x + 2$
Degree 3 ✓

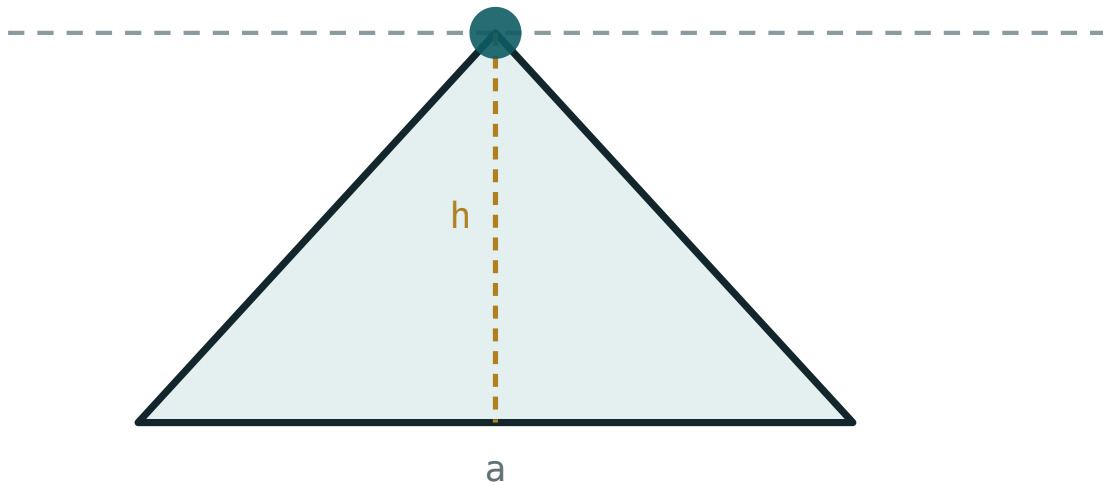
ANSWER

$$x^3 - x^2 - 5x + 2$$

03 Interactive model

Draw the rectangle for $(x + 2)(x + 3)$ with its four parts labelled; the expansion is read off the picture before any rule is stated.

Expanding as an area

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**
04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply out	Ko'paytirish	Умножение
To expand	Yoyish	Раскрыть
Binomial	Ikki had	Двучлен
Trinomial	Uch had	Трёхчлен
Every term	Har bir had	Каждый член
Area diagram	Yuza chizmasi	Схема площадей
To collect	Ixchamlash	Приведение подобных
Product	Ko'paytma	Произведение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$3x(x - 4)$ equals:

$(a + b)(c + d)$ gives how many products?

$(x + 2)(x + 3)$ equals:

$(x - 3)(x + 3)$ equals:

A binomial times a trinomial gives:

The degree of a product is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $3x(x - 4)$

ANSWER $3x^2 - 12x$

2. $2a(a + 5)$

ANSWER $2a^2 + 10a$

3. $xy(x + y)$

ANSWER $x^2y + xy^2$

4. $(x + 2)(x + 3)$

ANSWER $x^2 + 5x + 6$

5. $(x + 1)(x + 4)$

ANSWER $x^2 + 5x + 4$

6. $(x - 3)(x + 3)$

ANSWER $x^2 - 9$

7. $(x + 5)(x - 5)$

ANSWER $x^2 - 25$

Medium · the standard question

7 PROBLEMS

1. $-2a(a^2 - 3a + 1)$

ANSWER $-2a^3 + 6a^2 - 2a$

2. $(x - 2)(x + 5)$

ANSWER $x^2 + 3x - 10$

3. $(2a - 1)(a + 4)$

ANSWER $2a^2 + 7a - 4$

4. $(3x + 2)(x - 1)$

ANSWER $3x^2 - x - 2$

5. $(x + 2)(x^2 - 3x + 1)$

ANSWER $x^3 - x^2 - 5x + 2$

6. $(a - b)(a + b)$

ANSWER $a^2 - b^2$

7. $(2x - 3)(2x + 3)$

ANSWER $4x^2 - 9$

Hard · stretch and reason

7 PROBLEMS

1. $(x + 1)(x + 2)(x + 3)$

ANSWER $x^3 + 6x^2 + 11x + 6$

2. $(a + b)^2$

ANSWER $a^2 + 2ab + b^2$

3. $(a - b)^2$

ANSWER $a^2 - 2ab + b^2$

4. $(2x - 1)(x^2 + x - 3)$

ANSWER $2x^3 + x^2 - 7x + 3$

5. A rectangle $(x + 3)$ by $(x + 5)$: its area

ANSWER $x^2 + 8x + 15$

6. Its perimeter

ANSWER $4x + 16$

7. $(x + a)(x + b)$ in general

ANSWER $x^2 + (a + b)x + ab$

07 Homework**Homework — 5 tasks**

Count the products you expect before expanding, and again afterwards.

1. Expand $4x(2x - 3)$ and $-3a(a^2 + 2a - 5)$.
2. Expand $(x + 4)(x + 6)$ and $(x - 4)(x + 6)$.
3. Expand $(3a + 2)(2a - 5)$.
4. Expand $(x - 1)(x^2 + 2x + 3)$.
5. A rectangle has sides $x + 2$ and $x + 7$. Find its area and perimeter.

Division of polynomials

Dividing by a monomial — term by term, and only when it goes exactly.

LESSON 75–76

2 × 40 MIN

MATEMATIKA 7, §35

S8 2.3

LESSON CLOCK

13'

23'

22'

18'

4'

Term by term	13 min
When it works	23 min
Reversing a multiplication	22 min
Simplifying fractions	18 min
Homework	4 min

LEARNING OBJECTIVES

- Divide a polynomial by a monomial, term by term.
- Decide when the quotient is again a polynomial.
- Use division to reverse a multiplication.
- Simplify an algebraic fraction whose denominator is a monomial.

TEXTBOOK

National	pp. 212–217
Cambridge	Stage 8, pp. 18–24
Workbook	Exercise 2.3

01

Explanation

Term by term

$$\frac{a + b + c}{k} = \frac{a}{k} + \frac{b}{k} + \frac{c}{k}$$

Divide each term of the polynomial by the monomial.

QUOTIENT	TERM BY TERM	ANSWER
$(6x^2 + 9x) \div 3x$	$\frac{6x^2}{3x} + \frac{9x}{3x}$	$2x + 3$
$(8a^3 - 4a^2) \div 4a^2$	$2a - 1$	$2a - 1$
$(10x^3y - 15x^2y^2) \div 5x^2y$	$2x - 3y$	$2x - 3y$

EVERY TERM IS DIVIDED, INCLUDING THE LAST

$(8a^3 - 4a^2) \div 4a^2$ is $2a - 1$, not $2a$. The second term gives -1 , and dropping it is the standard slip.

When it works

The quotient is a polynomial only if the monomial divides **every** term exactly — the coefficients and each letter.

DIVISION	POLYNOMIAL?	WHY
$(6x^2 + 9x) \div 3x$	yes	$3x$ divides both terms
$(6x^2 + 9x) \div 3x^2$	no	$3x^2$ does not divide $9x$
$(4a + 6b) \div 2$	yes	2 divides both terms
$(4a + 5b) \div 2$	no	2 does not divide $5b$

THE DIVISOR MUST BE A COMMON FACTOR

Finding the highest common factor of all the terms tells you at once what the polynomial can be divided by — and that is exactly what factorising will use two lessons from now.

Reversing a multiplication

Division undoes multiplication, so every division can be checked by multiplying back.

DIVISION	ANSWER	CHECK
$(6x^2 + 9x) \div 3x$	$2x + 3$	$3x(2x + 3) = 6x^2 + 9x$ ✓
$(10a^2 - 5a) \div 5a$	$2a - 1$	$5a(2a - 1) = 10a^2 - 5a$ ✓

ONE MULTIPLICATION CHECKS THE WHOLE DIVISION

It takes ten seconds and catches every slip, including the dropped last term. There is no faster check in algebra.

Simplifying fractions

An algebraic fraction is simplified by cancelling the common factors of numerator and denominator.

FRACTION	SIMPLIFIED
$\frac{12a^3}{4a}$	$3a^2$
$\frac{6x^2 + 9x}{3x}$	$2x + 3$
$\frac{5ab}{10a^2}$	$\frac{b}{2a}$

CANCEL FACTORS, NEVER TERMS

In $\frac{6x^2 + 9x}{3x}$ the whole numerator is divided. Cancelling the x of $9x$ against the denominator alone, and leaving the $6x^2$ untouched, is wrong — the division applies to every term or to none.

02

Worked examples

EXAMPLE 1 Compute $(6x^2 + 9x) \div 3x$.

1

$$\frac{6x^2}{3x} = 2x$$

2

$$\frac{9x}{3x} = 3$$

3

$$= 2x + 3$$

4

Check: $3x(2x + 3) = 6x^2 + 9x$ ✓

ANSWER

$$2x + 3$$

EXAMPLE 2 Compute $(10x^3y - 15x^2y^2) \div 5x^2y$.

① First term: $\frac{10x^3y}{5x^2y} = 2x$.

② Second: $\frac{-15x^2y^2}{5x^2y} = -3y$.

③ $= 2x - 3y$

④ Check: $5x^2y(2x - 3y) = 10x^3y - 15x^2y^2$ ✓

ANSWER

$$2x - 3y$$

EXAMPLE 3 Is $(6x^2 + 9x) \div 3x^2$ a polynomial?

① First term: $\frac{6x^2}{3x^2} = 2$ ✓

② Second: $\frac{9x}{3x^2} = \frac{3}{x}$.

③ That is not a monomial.

④ No — the quotient is not a polynomial.

ANSWER

No

03 Interactive model

Ask the class to check every division by multiplying back, and mark only the checks for one lesson; the accuracy of the divisions rises immediately.

Cancelling common factors

$$\frac{6x^2 + 9x}{3x}$$

① $\frac{6x^2}{3x} = 2x$

The
first
term.

② $\frac{9x}{3x} = 3$

The
second
— it
does
not
vanish.

③ $2x + 3$

Check:

$$\begin{aligned} &3x(2x \\ &+ 3) \\ &= \\ &6x^2 \\ &+ 9x \\ &\checkmark \end{aligned}$$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Division	Bo'lish	Деление
Quotient	Bo'linma	Частное
Term by term	Hadma-had	Почленно
Exactly divisible	Butun bo'linadi	Делится нацело
Common factor	Umumiy ko'paytuvchi	Общий множитель
To reverse	Teskari amal	Обратное действие
Algebraic fraction	Algebraik kasr	Алгебраическая дробь
Check	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To divide a polynomial by a monomial:

divide the first term

divide every term

multiply

cancel one letter

$(8a^3 - 4a^2) \div 4a^2$ equals:

$2a$

$2a - 1$

$2a^3$

$2a - 4$

$(6x^2 + 9x) \div 3x$ equals:

$2x$

$2x + 3$

$2x + 9$

$2x^2 + 3$

$(6x^2 + 9x) \div 3x^2$ is:

a polynomial

not a polynomial

a monomial

zero

A division is checked by:

dividing again

multiplying back

estimating

nothing

In an algebraic fraction you may cancel:

terms

factors

anything

signs

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(6x^2 + 9x) \div 3x$

ANSWER $2x + 3$

2. $(8a^3 - 4a^2) \div 4a^2$

ANSWER $2a - 1$

3. $(4a + 6b) \div 2$

ANSWER $2a + 3b$

4.
$$\frac{12a^3}{4a}$$

ANSWER $3a^2$

5. $(10a^2 - 5a) \div 5a$

ANSWER $2a - 1$

6. $(x^2 + x) \div x$

ANSWER $x + 1$

7. $(9y^3 + 3y) \div 3y$

ANSWER $3y^2 + 1$ **Medium · the standard question**

7 PROBLEMS

1. $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER $2x - 3y$

2. $(12a^2b + 8ab^2) \div 4ab$

ANSWER $3a + 2b$

3. $\frac{5ab}{10a^2}$

ANSWER $\frac{b}{2a}$

4. Is $(6x^2 + 9x) \div 3x^2$ a polynomial?

ANSWER No

5. Is $(4a + 5b) \div 2$ a polynomial?

ANSWER No

6. Check $2x + 3$ by multiplying by $3x$

ANSWER $6x^2 + 9x$

7. $(a^3 - a^2) \div a^2$

ANSWER $a - 1$

Hard · stretch and reason

7 PROBLEMS

1. $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER $3x^2y - 2xy^2 + 1$

2.
$$\frac{4a^2b - 6ab^2}{2ab}$$

ANSWER $2a - 3b$

3. What must $3x$ be multiplied by to give $12x^3 - 9x^2$?

ANSWER $4x^2 - 3x$

4.
$$\frac{x^2 - x}{x}$$

ANSWER $x - 1$

5. The HCF of $6x^2$ and $9x$

ANSWER $3x$

6. Divide $8a^3b - 12a^2b^2 + 4ab$ by its HCF

ANSWER $2a^2 - 3ab + 1$

7. Why must the divisor be a common factor?

ANSWER Otherwise some term gives a fraction

07 Homework

Homework — 5 tasks

Check every division by multiplying the answer by the divisor.

1. Compute $(12x^2 - 8x) \div 4x$.

2. Compute $(9a^3b - 6a^2b^2) \div 3a^2b$.

3. Simplify $\frac{20m^3}{5m}$.

4. Is $(10x^2 + 4x) \div 4x^2$ a polynomial? Explain.

5. Find the HCF of $8a^2b$ and $12ab^2$, and divide their sum by it.

Factorising a polynomial

Taking out the common factor, and grouping when there is no single one.

LESSON 77–78

2 × 40 MIN

MATEMATIKA 7, §36

S8 2.4

LESSON CLOCK

9'

23'

22'

22'

4'

The idea	9 min
The common factor	23 min
Grouping	22 min
What it is for	22 min
Homework	4 min

LEARNING OBJECTIVES

- Find the highest common factor of the terms of a polynomial.
- Take that factor outside a bracket, and check by expanding.
- Factorise a four-term polynomial by grouping.
- Use a factorised form to evaluate an expression quickly.

TEXTBOOK

National	pp. 218–224
Cambridge	Stage 8, pp. 25–30
Workbook	Exercise 2.4

01

Explanation

The idea

Expanding turns a product into a sum. **Factorising** is the same road travelled backwards: it turns a sum into a product.

$$3x(2x + 3) = 6x^2 + 9x \rightarrow 6x^2 + 9x = 3x(2x + 3)$$

DIRECTION	START	FINISH
expanding	$3x(2x + 3)$	$6x^2 + 9x$
factorising	$6x^2 + 9x$	$3x(2x + 3)$

THE CHECK IS FREE

Every factorisation can be checked by expanding it again. If the expansion does not return the original polynomial exactly, the factorisation is wrong — there is nothing to argue about.

The common factor

Take the highest common factor of **all** the terms outside the bracket. For the numbers that is their HCF; for each letter it is the lowest power that appears in every term.

POLYNOMIAL	HCF	FACTORISED
$6x^2 + 9x$	$3x$	$3x(2x + 3)$
$8a^3 - 4a^2$	$4a^2$	$4a^2(2a - 1)$
$10x^3y - 15x^2y^2$	$5x^2y$	$5x^2y(2x - 3y)$
$12a^2b + 8ab^2 - 4ab$	$4ab$	$4ab(3a + 2b - 1)$

A TERM THAT IS ENTIRELY THE FACTOR LEAVES A 1

In $12a^2b + 8ab^2 - 4ab$ the last term is $-4ab$, which is the factor itself; inside the bracket it becomes -1 , not nothing. Losing that 1 is the commonest error of the whole topic, and expanding back catches it at once.

Grouping

When the four terms have no common factor between all of them, split them into two pairs that do, and a common bracket appears.

$$ax + ay + bx + by = a(x + y) + b(x + y) = (x + y)(a + b)$$

STEP	WORKING
the polynomial	$2x^2 + 6x + ax + 3a$
pair them	$(2x^2 + 6x) + (ax + 3a)$
factorise each pair	$2x(x + 3) + a(x + 3)$
the bracket is common	$(x + 3)(2x + a)$

A MINUS IN FRONT OF THE SECOND PAIR CHANGES BOTH SIGNS

$x^2 - 3x - 2x + 6$ pairs as $x(x - 3) - 2(x - 3)$, giving $(x - 3)(x - 2)$. Writing $-2(x + 3)$ there is a sign slip, and the two brackets then fail to match.

What it is for

A product is easier to evaluate, easier to cancel and — next year — the only way to solve an equation.

USE	EXAMPLE	WHY THE PRODUCT HELPS
evaluating	$37 \cdot 24 + 37 \cdot 76$	$37(24 + 76) = 3700$
cancelling	$\frac{6x^2 + 9x}{3x}$	$\frac{3x(2x + 3)}{3x} = 2x + 3$
divisibility	$n^2 + n$	$n(n + 1)$

WHY $n^2 + n$ IS ALWAYS EVEN

Factorised it is $n(n + 1)$ — the product of two consecutive whole numbers, one of which must be even. In the unfactorised form the fact is invisible; that is the whole value of a product.

02 Worked examples

EXAMPLE 1 Factorise $12a^2b + 8ab^2 - 4ab$.

- ① HCF of 12, 8, 4 is 4.
- ② Every term has a and b , so the HCF is $4ab$.
- ③ $= 4ab(3a + 2b - 1)$
The last term leaves -1 .
- ④ Check: $4ab(3a + 2b - 1) = 12a^2b + 8ab^2 - 4ab$ ✓

ANSWER

$$4ab(3a + 2b - 1)$$

EXAMPLE 2 Factorise $2x^2 + 6x + ax + 3a$ by grouping.

$$(1) \quad (2x^2 + 6x) + (ax + 3a)$$

$$(2) \quad 2x(x + 3) + a(x + 3)$$

The same bracket in both.

$$(3) \quad = (x + 3)(2x + a)$$

$$(4) \quad \text{Check: } (x + 3)(2x + a) = 2x^2 + ax + 6x + 3a \quad \checkmark$$

ANSWER

$$(x + 3)(2x + a)$$

EXAMPLE 3 Evaluate $37 \cdot 24 + 37 \cdot 76$ without a calculator.

$$(1) \quad \text{Both terms contain 37.}$$

$$(2) \quad = 37(24 + 76)$$

$$(3) \quad = 37 \cdot 100$$

$$(4) \quad = 3700$$

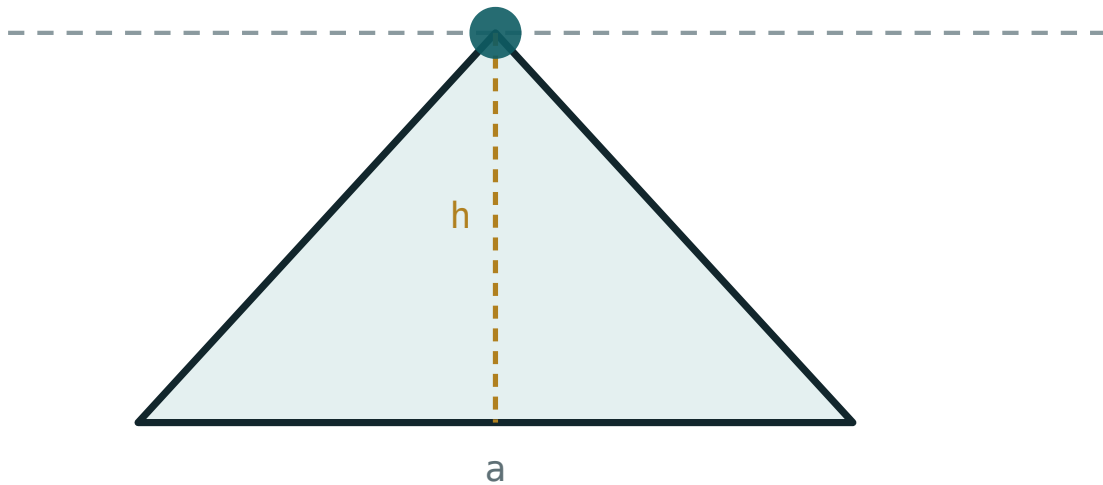
ANSWER

$$3700$$

03 Interactive model

Give the class the factorised answers first and ask them to expand; when they then meet the sums, the factor is already familiar and the missing 1 stops appearing.

A common factor as one side of a rectangle

base a **220**height h **120**area $\frac{1}{2}ah$ **13200**perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Factor	Ko'paytuvchi	Множитель
Highest common factor	Eng katta umumiy bo'luvchi	Наибольший общий делитель
To take outside the bracket	Qavsdan tashqariga chiqarish	Вынести за скобку
Grouping	Guruhlash	Группировка
Product	Ko'paytma	Произведение
Sum	Yig'indi	Сумма
Check by expanding	Qavsni ochib tekshirish	Проверка раскрытием

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Factorising turns a sum into:

a difference

a product

a quotient

a power

$6x^2 + 9x$ factorises to:

$3x(2x + 3)$

$3x(2x + 9)$

$x(6x + 9)$

$3(2x^2 + 3x)$

In $12a^2b + 8ab^2 - 4ab$ the last term leaves:

0

-1

-4ab

1

Grouping is used when:

there are two terms

no factor is common to all

the terms are like

there is a power

$ax + ay + bx + by$ equals:

$(x + y)(a + b)$

$(a + x)(b + y)$

$ab(x + y)$

$(a + b)^2$

A factorisation is checked by:

dividing

expanding

substituting 0

nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $4x + 8$

ANSWER $4(x + 2)$

2. $6x^2 + 9x$

ANSWER $3x(2x + 3)$

3. $8a^3 - 4a^2$

ANSWER $4a^2(2a - 1)$

4. $5x - 15$

ANSWER $5(x - 3)$

5. $x^2 + x$

ANSWER $x(x + 1)$

6. $9y^3 + 3y$

ANSWER $3y(3y^2 + 1)$

7. $10ab - 5a$

ANSWER $5a(2b - 1)$

Medium · the standard question**7 PROBLEMS**

1. $10x^3y - 15x^2y^2$

ANSWER $5x^2y(2x - 3y)$

2. $12a^2b + 8ab^2 - 4ab$

ANSWER $4ab(3a + 2b - 1)$

3. $ax + ay + bx + by$

ANSWER $(x + y)(a + b)$

4. $2x^2 + 6x + ax + 3a$

ANSWER $(x + 3)(2x + a)$

5. $x^2 - 3x - 2x + 6$

ANSWER $(x - 3)(x - 2)$

6. $37 \cdot 24 + 37 \cdot 76$

ANSWER 3700

7. $14m^2n - 21mn^2$

ANSWER $7mn(2m - 3n)$

Hard · stretch and reason

7 PROBLEMS

1. $15x^4y^2 - 10x^3y^3 + 5x^2y$

ANSWER $5x^2y(3x^2y - 2xy^2 + 1)$

2. $3a(x - 2) + 5(x - 2)$

ANSWER $(x - 2)(3a + 5)$

3. $x(a - b) - y(b - a)$

ANSWER $(a - b)(x + y)$

4. Show that $n^2 + n$ is always even

ANSWER $n(n + 1)$ — one of them is even

5. $56 \cdot 19 + 56 \cdot 81$

ANSWER 5600

6. Simplify $\frac{4a^2b - 6ab^2}{2ab}$ by factorising first

ANSWER $2a - 3b$

7. Factorise $x^3 + x^2 + x + 1$

ANSWER $(x + 1)(x^2 + 1)$

07 Homework**Homework — 5 tasks**

Expand every answer back; a factorisation that does not return the original is not an answer.

- Factorise $12x^2 - 8x$ and $9a^3b - 6a^2b^2$.
- Factorise $6m^2n + 9mn^2 - 3mn$, and say what the last term leaves.
- Factorise $3x^2 + 12x + ax + 4a$ by grouping.
- Evaluate $48 \cdot 37 + 48 \cdot 63$ by factorising.

5. Explain, using a factorisation, why $n^2 - n$ is always even.

Control work 5, and work on the mistakes

Expressions, powers, monomials and polynomials in one paper — Quarter II closed.

LESSON 79–80

2 × 40 MIN

MATEMATIKA 7, NAZORAT ISHI 5

S8 2 REVIEW

LESSON CLOCK

3

40'

12'

20'

5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Apply the laws of indices without confusing them with the laws of arithmetic.
- Add, subtract, multiply and divide polynomials accurately under time.
- Factorise by the common factor and by grouping.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 152–224
Cambridge	Stage 8, pp. 12–30
Workbook	Control paper M5

01

Explanation

The paper — 25 marks, 40 minutes

Q	TASK	MARKS	FROM
1	Simplify $a^5 \cdot a^3$, $a^8 \div a^3$, $(a^2)^4$ and a^0	4	L63–64
2	Multiply the monomials $3x^2y$ and $-4xy^3$	3	L66–67
3	Collect $5a^2 - 3a + 2 - 2a^2 + 7a - 5$	4	L69–70
4	Expand $(x + 4)(x - 3)$ and $-3a(a^2 + 2a - 5)$	5	L73–74
5	Divide $(10x^3y - 15x^2y^2) \div 5x^2y$	4	L75–76
6	Factorise $12a^2b + 8ab^2 - 4ab$ and $2x^2 + 6x + ax + 3a$	5	L77–78

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for $a^0 = 1$ alone; Q2 one for the sign; Q4 two for the four products of the binomial pair and one for the sign of the last term; Q6 one for the -1 left by the third term and two for the common bracket in the grouping.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
indices multiplied	$a^5 \cdot a^3 = a^{15}$	a^8 — add them
zero index misread	$a^0 = 0$	$a^0 = 1$
sign of a product	$3x^2y \cdot (-4xy^3) = 12x^3y^4$	$-12x^3y^4$
unlike terms collected	$5a^2 - 3a = 2a$	they are not like
only two products	$(x + 4)(x - 3) = x^2 - 12$	all four, then collect
last term of a division dropped	$2x$	$2x - 3y$
the 1 lost when factorising	$4ab(3a + 2b)$	$4ab(3a + 2b - 1)$

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Quarter II as one map

BLOCK	THE SENTENCE
expressions	a letter stands for a number, so the laws of arithmetic still hold
powers	an index counts the factors, so multiplying adds the indices
monomials	one product of numbers and letters, with a degree
polynomials	a sum of monomials; only like terms may be collected
multiplication	every term of the first by every term of the second
division	term by term, and only when the divisor is a common factor
factorising	the common factor outside, or grouping in pairs

LOOKING FORWARD

Quarter III opens with geometry again — parallel lines, the angle sum of a triangle and right-angled triangles — and returns to algebra with the abridged multiplication formulae, which are the products of Q4 learned by heart.

02 Worked examples

EXAMPLE 1 Model answer, Q1: the four index tasks.

① $a^5 \cdot a^3 = a^8$
Add the indices.

② $a^8 \div a^3 = a^5$
Subtract them.

③ $(a^2)^4 = a^8$
Multiply them.

④ $a^0 = 1$
For every $a \neq 0$.

ANSWER

$a^8, a^5, a^8, 1$

EXAMPLE 2 Model answer, Q4: expand $(x + 4)(x - 3)$.

$$1 \quad x \cdot x = x^2$$

$$2 \quad x \cdot (-3) = -3x \text{ and } 4 \cdot x = 4x.$$

$$3 \quad 4 \cdot (-3) = -12$$

The sign is the mark.

$$4 \quad = x^2 + x - 12$$

ANSWER

$$x^2 + x - 12$$

EXAMPLE 3 Model answer, Q6: factorise $2x^2 + 6x + ax + 3a$.

$$1 \quad (2x^2 + 6x) + (ax + 3a)$$

$$2 \quad 2x(x + 3) + a(x + 3)$$

$$3 \quad = (x + 3)(2x + a)$$

$$4 \quad \text{Check by expanding: } 2x^2 + ax + 6x + 3a \checkmark$$

ANSWER

$$(x + 3)(2x + a)$$

03 Interactive model

Hand back Q4 and Q6 together and ask which is which: expanding and factorising are one skill in two directions, and seeing them side by side fixes it.

Quarter II in twelve questions

 $a^5 \cdot a^3$ equals:

a^{15}

a^8

a^2

$2a^8$

 $a^8 \div a^3$ equals:

a^5

a^{11}

a^2

a^{24}

 $(a^2)^4$ equals:

a^6

a^8

a^{16}

$2a^4$

a^0 equals:

0

1

a

undefined

The degree of $3x^2y$ is:

2

3

5

1

$3x^2y \cdot (-4xy^3)$ equals:

$12x^3y^4$

$-12x^3y^4$

$-12x^2y^3$

$-7x^3y^4$

Like terms differ only in:

the letters

the coefficient

the degree

the sign of the letter

$5a^2 - 3a + 2 - 2a^2 + 7a - 5$ equals:

$$3a^2 + 4a - 3$$

$$7a^2 + 4a - 3$$

$$3a^2 - 4a + 3$$

$$3a^2 + 10a - 3$$

$(x + 4)(x - 3)$ equals:

$$x^2 - 12$$

$$x^2 + x - 12$$

$$x^2 + 7x - 12$$

$$x^2 - x - 12$$

$(8a^3 - 4a^2) \div 4a^2$ equals:

$$2a$$

$$2a - 1$$

$$2a - 4$$

$$2a^3$$

$12a^2b + 8ab^2 - 4ab$ factorises to:

$$4ab(3a + 2b)$$

$$4ab(3a + 2b - 1)$$

$$ab(12a + 8b - 4)$$

$$4(3a^2b + 2ab^2 - ab)$$

A factorisation is checked by:

dividing

expanding

substituting

nothing

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Power	Daraja	Степень
Monomial	Bir had	Одночлен
Polynomial	Ko'phad	Многочлен
Like terms	O'xshash hadlar	Подобные члены
To expand	Qavsni ochish	Раскрыть скобки
To factorise	Ko'paytuvchilarga ajratish	Разложить на множители
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is answered with:

the laws of arithmetic

the laws of indices

a calculator

estimation

The sign in Q2 comes from:

the letters

one negative factor

the degree

the coefficients only

In Q3 the terms $5a^2$ and $-3a$ are:

like

not like

equal

opposite

Q4 needs how many products for the binomial pair?

2

3

4

6

Q5 is wrong if:

the first term is divided

the last term is dropped

the check is done

the answer is a polynomial

The commonest slip in Q6 is:

the wrong HCF

the lost 1

too many brackets

expanding

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $a^5 \cdot a^3$

ANSWER a^8

2. $a^8 \div a^3$

ANSWER a^5

3. $(a^2)^4$

ANSWER a^8

4. a^0 , for $a \neq 0$

ANSWER 1

5. $3x^2y \cdot (-4xy^3)$

ANSWER $-12x^3y^4$

6. $5a^2 - 2a^2$

ANSWER $3a^2$

7. $(x^2 + x) \div x$

ANSWER $x + 1$ **Medium · the standard question**

7 PROBLEMS

1. $5a^2 - 3a + 2 - 2a^2 + 7a - 5$

ANSWER $3a^2 + 4a - 3$

2. $(x + 4)(x - 3)$

ANSWER $x^2 + x - 12$

3. $-3a(a^2 + 2a - 5)$

ANSWER $-3a^3 - 6a^2 + 15a$

4. $(10x^3y - 15x^2y^2) \div 5x^2y$

ANSWER $2x - 3y$

5. $12a^2b + 8ab^2 - 4ab$

ANSWER $4ab(3a + 2b - 1)$

6. $2x^2 + 6x + ax + 3a$

ANSWER $(x + 3)(2x + a)$

7. The degree of $3x^2y$

ANSWER 3

Hard · stretch and reason**7 PROBLEMS**

1. $(2x - 1)(x^2 + x - 3)$

ANSWER $2x^3 + x^2 - 7x + 3$

2. $(15x^4y^2 - 10x^3y^3 + 5x^2y) \div 5x^2y$

ANSWER $3x^2y - 2xy^2 + 1$

3. $x(a - b) - y(b - a)$

ANSWER $(a - b)(x + y)$

4. Simplify $\frac{6x^2 + 9x}{3x}$

ANSWER $2x + 3$

5. Expand and collect $(a + b)^2 - (a - b)^2$

ANSWER $4ab$

6. $48 \cdot 37 + 48 \cdot 63$

ANSWER 4800

7. Why is $n^2 + n$ always even?

ANSWER $n(n + 1)$ — one factor is even

07 Homework**Homework — 5 tasks**

Rewrite every question you lost a mark on in full, from the first line.

1. Rewrite in full every question on which you lost a mark.
2. Write out the four laws of indices with one example each.
3. Expand $(2x - 5)(x + 3)$ and check by substituting $x = 1$.
4. Factorise $9m^2n + 12mn^2 - 3mn$ and expand your answer back.

5. Write one sentence for each of the seven blocks on the Quarter II map.